Purpose-Based Spectral Analysis: Domain-Constrained Generative Identification Through Oscillatory Resonance in Bounded Phase Space

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Abstract

Mass spectrometric identification conventionally requires $O(N)$ comparisons against a spectral library of N compounds, with each comparison evaluating an algorithmic similarity function (Tanimoto coefficient, cosine similarity, dot product). We demonstrate that both the $O(N)$ scaling and the algorithmic comparison are artefacts of an extrinsic representation: when molecular identity is encoded in an intrinsic oscillatory coordinate system, comparison becomes resonance and search becomes navigation. Starting from the Bounded Phase Space Law—an axiom asserting that every observable system occupies a finite-volume region in phase space—we derive that comparison between bounded oscillatory systems is not computation but physical resonance: two systems either share oscillatory structure (co-locate in phase space) or they do not, requiring no similarity function. We establish dual oscillatory paths (spectral frequencies and droplet surface modes) that independently converge to identical S-entropy coordinates, providing algorithm-free cross-validation. We then formalize the role of domain-specific intelligence: a Purpose function $\mathcal{P}\colon \mathcal{C} \to 2^{\mathcal{S}}$ maps analytical context (sample type, instrument, biological question) to constrained subregions of S-entropy phase space, determining *which* molecular trajectories are generated from the otherwise vast space of possibilities. This purpose-based constraint reduces the effective search from 3^k possible addresses to r domain-relevant prefixes, where typically $r/3^k < 0.05$. We prove that selective generation achieves identification in $O(r \cdot k)$ operations—independent of both database size N and total phase space volume—while preserving identification accuracy whenever domain constraints are correct. The Molecular Maxwell Demon formalism quantifies this reduction: domain knowledge is thermodynamic information that reduces search entropy by $\Delta S = k_{\mathrm{B}} \ln\left(3^k/|R|\right)$, at minimum energy cost satisfying the Landauer bound. We validate across three domains (metabolomics, glycomics, proteomics) demonstrating >95% computation reduction with zero accuracy loss when domain constraints contain the target. The central result: when the purpose determines the phase space region and comparison is oscillatory resonance, identification becomes a physical process rather than a computational one.

Keywords: oscillatory resonance, purpose-based identification, domain constraint, Molecular Maxwell Demon, S-entropy coordinates, bounded phase space, selective generation, mass spectrometry

Part I
Foundations

1 Introduction

1.1 The Identification Problem

Mass spectrometric identification of small molecules and metabolites proceeds, in essentially all existing implementations, through a three-stage pipeline: (i) acquire a mass spectrum from the analyte of interest, (ii) compare the acquired spectrum against every entry in a spectral library of N reference compounds using an algorithmic similarity func-

tion, and (iii) rank candidate matches by their computed similarity score [Stein and Scott, 1994, Gross, 2017]. This pipeline, despite decades of optimization, suffers from a double inefficiency that is structural rather than implementational.

The first inefficiency is the $O(N)$ scaling of the comparison step. For every unknown spectrum, the system must evaluate a similarity function against each of the N library entries, where N ranges from $\sim 10^4$ for curated metabolomics databases [Wishart et al., 2022] to $\sim 10^8$ for comprehensive chemical databases [Kim et al., 2023]. The vast majority of these comparisons are irrelevant—a human serum metabolite will not match an industrial polymer, yet the algorithm evaluates both. Approximate nearest-neighbor methods [Indyk and Motwani, 1998] reduce the constant factor but preserve the fundamental scaling.

The second inefficiency is subtler and more consequential: each of the $O(N)$ comparisons evaluates an *algorithmic* similarity function. Whether the function is a Tanimoto coefficient over molecular fingerprints [Bajusz et al., 2015, Willett et al., 1998], a cosine similarity over spectral vectors [Stein and Scott, 1994], or an extended-connectivity fingerprint comparison [Rogers and Hahn, 2010, Riniker and Landrum, 2013], the comparison is a *computation* performed on *extrinsic* representations—binary fingerprints, intensity vectors, or graph features that are external to the physical identity of the molecule. The molecule does not know its Tanimoto coefficient; the coefficient is a property of the representation, not of the molecule.

This observation leads to a deeper question: *why is comparison treated as computation at all?* If molecular identity is a physical property, and if this property can be encoded in an intrinsic coordinate system, then comparison between molecules should be a physical relationship—one that can be determined by inspection of coordinates rather than computation of similarity functions.

1.2 Two Claims

This paper advances two claims that, taken together, eliminate both inefficiencies simultaneously.

Claim 1 (Comparison as Resonance). Comparison between bounded oscillatory systems is resonance, not computation. When molecular identity is encoded in oscillatory coordinates—frequency spectra rather than fingerprints—two molecules are "similar" if and only if their oscillatory modes overlap in a well-defined sense. This overlap is a structural property of the coordinate system, not a computation performed on it. Determining whether two addresses share a common prefix requires only $O(k)$ trit comparisons, where k is the resolution depth, independent of molecular complexity, fingerprint length, or database size.

Claim 2 (Purpose as Phase Space Constraint). Domain knowledge constrains the accessible phase space *before* generation, eliminating most comparisons entirely. An analyst performing metabolomics on human serum does not need to compare against polymer databases, glycan libraries, or atmospheric pollutant spectra. This prior knowledge—sample type, organism, disease state, instrument configuration—defines a subregion of phase space within which all relevant molecular identities reside. By generating trajectories only within this subregion, the vast majority of the 3^k possible phase-space cells are never visited.

These two claims are not independent. Resonance-based comparison (Claim 1) ensures that each individual comparison costs $O(k)$ rather than $O(d)$ where d is the fingerprint dimension. Purpose-based constraint (Claim 2) ensures that only $r \ll 3^k$ comparisons are needed. Together, they yield identification in $O(r \cdot k)$—independent of both database size N and phase-space volume 3^k.

1.3 The Purpose Function

In analytical chemistry, the analyst always knows something before measuring. The sample arrived from a clinical laboratory (constraining the expected metabolome), it was prepared with a specific protocol (constraining the mass range and ionization state), it was injected into a particular instrument (constraining the resolution and mass accuracy), and the analysis serves a specific biological question (constraining which identifications are meaningful). This prior knowledge is not merely useful context—it is *thermodynamic information* that reduces the entropy of the search space.

We formalize this insight through a *Purpose function* $\mathcal{P}\colon \mathcal{C} \to 2^{\mathcal{S}}$ that maps a set of domain constraints (the analytical context $\mathcal{C}$) to a subregion of the S-entropy phase space $\mathcal{S}$. The Purpose function determines *which* molecular trajectories are generated from the otherwise vast space of possibilities, implementing what we call a *Molecular Maxwell Demon* at the meta level: a thermodynamic filter that operates not on individual ion states but on the *choice of which ions to consider*.

1.4 Scope and Contributions

This paper makes the following specific contributions:

(i) **Oscillatory foundations from BPSL.** We derive the necessity of oscillatory dynamics from the Bounded Phase Space Law (BPSL), establishing that all bounded observable systems must exhibit periodic motion with discrete frequency spectra (Section 2).

(ii) **Comparison as resonance.** We prove that comparison between bounded oscillatory systems encoded in S-entropy coordinates reduces to checking address co-location, which is a resonance condition, not a computation (Section 4).

(iii) **Dual oscillatory path convergence.** We establish that two independent oscillatory representations of molecular identity—spectral frequencies (the ion path) and surface wave modes (the droplet path)—converge to identical S-entropy coordinates (Section 5).

(iv) **Purpose function formalization.** We formalize domain-specific intelligence as a Purpose function mapping analytical context to phase-space subregions, and prove monotonic contraction under additional constraints (Section 7).

(v) **Selective generation theorem.** We prove that identification through purpose-constrained generation achieves $O(r \cdot k)$ operations, independent of database size N and total phase-space volume 3^k (Section 8).

(vi) **Molecular Maxwell Demon.** We quantify the entropy reduction from purpose-based filtering and prove it satisfies the Landauer bound, establishing domain knowledge as thermodynamic information (Section 6).

(vii) **Domain validation.** We validate across three analytical domains—metabolomics, glycomics, and proteomics—demonstrating >95% computation reduction with zero accuracy loss under correct domain constraints (Section 10).

1.5 Paper Structure

The paper is organized into five parts. Part I (Sections 1–3) establishes the mathematical foundations: the Bounded Phase Space Law, the triple equivalence theorem unique to this work, and the S-entropy coordinate system with its ternary addressing scheme. Part II (Sections 4–6) develops the oscillatory comparison framework, proving that comparison is resonance, establishing dual-path convergence, and formalizing the Molecular Maxwell Demon. Part III (Sections 7–9) formalizes purpose as a phase-space constraint, derives the selective generation theorem, and computes domain-specific reduction bounds. Part IV (Sections 10–8.5) presents experimental validation. Part V (Sections 12–13) discusses implications and concludes.

2 Bounded Phase Space and the Triple Equivalence

2.1 The Axiom

We begin from a single axiom that serves as the foundation for all subsequent derivations. This paper is self-contained; no reference to prior work is required to follow the logical development.

Axiom 2.1 (Bounded Phase Space Law (BPSL))**.** Every observable system occupies a connected region of finite volume in its phase space $\Gamma \subset \mathbb{R}^{2n}$. That is, for any observable system with generalized coordinates $(q_1, \ldots, q_n)$ and conjugate momenta $(p_1, \ldots, p_n)$, the accessible phase-space volume satisfies:

$$V_\Gamma = \int_\Gamma \prod_{i=1}^{n} dq_i \, dp_i < \infty. \tag{1}$$

Moreover, Γ is a connected, compact subset of $\mathbb{R}^{2n}$.

The BPSL is an observational axiom, not a derived theorem. It encodes the empirical fact that every system we can observe, measure, or interact with occupies a finite region of phase space. An electron in an atom has finite energy (bounded momentum) and finite spatial extent (bounded position). A molecule in a mass spectrometer has finite mass, finite charge, finite kinetic energy. The axiom excludes only systems of infinite spatial extent or infinite energy—precisely those systems that are unobservable.

Remark 2.2. The compactness requirement in Theorem 2.1 is physically motivated: every measurement has finite precision, every detector has finite range, and every experimental system is spatially bounded. Mathematically, compactness of Γ ensures the existence of maxima and minima of continuous functions on Γ (extreme value theorem) and the validity of Poincaré recurrence (below).

2.2 Poincaré Recurrence

The first consequence of the BPSL is that trajectories in bounded phase space must eventually return to any neighborhood of their initial conditions.

Theorem 2.3 (Poincaré Recurrence)**.** *Let Γ be a compact phase-space region of finite Liouville measure, and let $\Phi^t \colon \Gamma \to \Gamma$ be a measure-preserving flow. Then for any measurable set $U \subset \Gamma$ with positive measure and for almost every point $x \in U$, there exists a sequence of times $\{t_n\}_{n=1}^{\infty}$ with $t_n \to \infty$ such that $\Phi^{t_n}(x) \in U$ for all n.*

Proof. Suppose for contradiction that there exists a measurable set $U \subset \Gamma$ with $\mu(U) > 0$ such that the set $W = \{x \in U : \Phi^t(x) \notin U \text{ for all } t > T\}$ has positive measure for some $T > 0$. Define the sequence of sets $W_n = \Phi^{nT}(W)$ for $n \geq 0$. Since Φ^t is measure-preserving, $\mu(W_n) = \mu(W) > 0$ for all n.

We claim that the sets $\{W_n\}_{n=0}^{\infty}$ are pairwise disjoint. For if $W_j \cap W_k \neq \emptyset$ with $j < k$, then there exists $x \in W$ such that $\Phi^{jT}(x) = \Phi^{kT}(y)$ for some $y \in W$, which gives $\Phi^{(k-j)T}(y) = \Phi^{jT}(x) \in W_j$. But $\Phi^{(k-j)T}(y)$ is a point in U that returns to a subset related to U after time $(k-j)T$, contradicting the definition of W.

But then $\mu(\Gamma) \geq \sum_{n=0}^{\infty} \mu(W_n) = \sum_{n=0}^{\infty} \mu(W) = \infty$, contradicting the finite volume condition $V_\Gamma < \infty$ of Theorem 2.1. Therefore $\mu(W) = 0$, and almost every point in U returns to U infinitely often. □

Remark 2.4. The essential ingredients are (i) finite phase-space volume (from the BPSL), (ii) measure-preservation (from Liouville's theorem for Hamiltonian systems, or more generally from the assumption of deterministic, invertible dynamics), and (iii) the pigeonhole principle: infinitely many iterates of a positive-measure set cannot all be disjoint in a finite-measure space. This is the original argument of Poincaré [1890].

2.3 Oscillatory Necessity

Poincaré recurrence establishes that trajectories return; we now strengthen this to show that trajectories must be *oscillatory*—that is, they must be decomposable into periodic modes.

Theorem 2.5 (Oscillatory Necessity)**.** *Let Γ be a compact phase-space region satisfying the BPSL, and let Φ^t be a measure-preserving flow on Γ. Then the trajectory $x(t) = \Phi^t(x_0)$ for almost every initial condition $x_0 \in \Gamma$ is quasi-periodic: there exist frequencies $\omega_1, \ldots, \omega_M$ and a function $F \colon \mathbb{T}^M \to \Gamma$ (where $\mathbb{T}^M$ is the M-dimensional torus) such that*

$$x(t) = F(\omega_1 t, \omega_2 t, \ldots, \omega_M t). \tag{2}$$

Proof. We proceed by eliminating all alternatives.

Case 1: Static trajectories. If $x(t) = x_0$ for all t, then the trajectory is a fixed point. Fixed points have zero measure in Γ (they form a set of dimension at most $2n-2$ in a $2n$-dimensional phase space). For almost every initial condition, the trajectory is non-static.

Case 2: Monotonic trajectories. A trajectory that moves monotonically in any coordinate q_i or p_i must eventually leave any compact region (since monotonic functions on $\mathbb{R}$ are unbounded on infinite time intervals). But Γ is compact, so the trajectory cannot leave Γ. Contradiction. Hence no trajectory can be asymptotically monotonic in any coordinate.

Case 3: Chaotic trajectories. By the Koopman–von Neumann theorem [Koopman, 1931], the time evolution operator $U^t f = f \circ \Phi^t$ acts as a unitary operator on $L^2(\Gamma, \mu)$. By the spectral theorem for unitary operators on separable Hilbert spaces, U^t admits a spectral decomposition. On a compact phase space, the spectrum of U^t is purely discrete (this follows from the compactness of Γ and the resulting discreteness of the Koopman spectrum for ergodic systems on compact manifolds). A discrete spectrum excludes continuous spectral components, which are the hallmark of mixing and chaotic dynamics [Arnold, 1963]. Therefore, the dynamics are not chaotic in the measure-theoretic sense.

Remaining case: Quasi-periodic motion. With static, monotonic, and chaotic alternatives eliminated, the trajectory must be quasi-periodic. By the Koopman spectral theorem, since the spectrum is discrete, there exist at most countably many frequencies $\{\omega_j\}$ such that every observable $f \in L^2(\Gamma, \mu)$ satisfies $f(\Phi^t(x)) = \sum_j c_j e^{i\omega_j t}$. Restricting to the M independent frequencies (those not related by integer linear combinations) gives the representation (2), where F is the embedding of the invariant torus into Γ. □

Corollary 2.6 (Mode Decomposition)**.** *Every bounded dynamical system admits a decomposition into $M \leq n$ independent oscillatory modes, where n is the number of degrees of freedom.*

Proof. The representation (2) involves M independent frequencies. Since the phase space is $2n$-dimensional and each mode contributes two dimensions (coordinate and momentum), we have $M \leq n$. Independence means the frequencies are rationally

independent: $\sum_{j=1}^{M} m_j\omega_j = 0$ with $m_j \in \mathbb{Z}$ implies $m_j = 0$ for all j. □

Corollary 2.7 (Frequency-Energy Identity)**.** *For each oscillatory mode j in a quantum-mechanical bounded system, the energy associated with that mode is*

$$E_j = \hbar\omega_j, \tag{3}$$

where ω_j is the angular frequency of mode j.

Proof. This follows from the correspondence principle applied to each independent mode. For a single harmonic mode of frequency ω_j, quantization gives energy levels $E_{j,n} = \hbar\omega_j(n + 1/2)$. The energy difference between adjacent levels is $\Delta E_j = \hbar\omega_j$. In the spectroscopic context, the fundamental relation $E = \hbar\omega$ connects observable transition frequencies to energy differences [Planck, 1901, Einstein, 1905, Bohr, 1913]. □

2.4 The Triple Equivalence

We now establish a result that is central to this paper and does not appear in other treatments: the *triple equivalence* between oscillatory, categorical, and partition-function descriptions of bounded systems. This equivalence is the bridge between the physics of oscillation and the mathematics of enumeration that underlies the entire framework.

Theorem 2.8 (Triple Equivalence)**.** *Let $\mathcal{B}$ be a bounded dynamical system with M independent degrees of freedom, each partitioned into n distinguishable states. Then the following three descriptions are equivalent:*

(1) ***Oscillatory:*** *$\mathcal{B}$ exhibits periodic motion with M modes, each sampling n phase states. The total number of distinct oscillatory configurations is $\Omega_{\text{osc}} = n^M$.*

(2) ***Categorical:*** *$\mathcal{B}$ undergoes sequential traversal of distinguishable states in an M-dimensional state space with n states per dimension. The total number of distinct categorical configurations is $\Omega_{\text{cat}} = n^M$.*

(3) ***Partition:*** *In the high-temperature limit, the canonical partition function of $\mathcal{B}$ with M independent subsystems, each having n equally accessible states, is $Z = n^M$.*

All three yield the same entropy:

$$S = k_{\text{B}} M \ln n. \tag{4}$$

Proof. We prove the equivalence by constructing explicit maps that form a closed cycle: Oscillatory → Categorical → Partition → Oscillatory.

Step 1: Oscillatory → Categorical. Consider mode j ($j = 1, \ldots, M$) oscillating at frequency ω_j. In one period $T_j = 2\pi/\omega_j$, the mode traverses a closed orbit in the (q_j, p_j) phase plane. Partition this orbit into n equal arcs, labeled $\{s_{j,1}, s_{j,2}, \ldots, s_{j,n}\}$. The time evolution of mode j visits these states sequentially: $s_{j,1} \to s_{j,2} \to \cdots \to s_{j,n} \to s_{j,1}$. This is precisely a categorical traversal of n states in dimension j.

The map ϕ_{OC} assigns to each oscillatory configuration (specifying which arc each mode occupies) the corresponding categorical state:

$$\phi_{\text{OC}} \colon (a_1, a_2, \ldots, a_M) \mapsto (s_{1,a_1}, s_{2,a_2}, \ldots, s_{M,a_M}),$$

where $a_j \in \{1, 2, \ldots, n\}$ labels the arc of mode j. This map is bijective: distinct arc labels map to distinct categorical states (injective), and every categorical state corresponds to some arc configuration (surjective). Hence $\Omega_{\text{osc}} = \Omega_{\text{cat}} = n^M$.

Step 2: Categorical → Partition. The canonical partition function at inverse temperature $\beta = 1/(k_{\text{B}}T)$ is

$$Z = \sum_{\text{states}} e^{-\beta E_{\text{state}}}. \tag{5}$$

For M independent subsystems, $Z = \prod_{j=1}^{M} Z_j$, where $Z_j = \sum_{i=1}^{n} e^{-\beta E_{j,i}}$.

In the high-temperature limit $\beta \to 0$, each Boltzmann factor approaches unity: $e^{-\beta E_{j,i}} \to 1$. Therefore $Z_j \to n$ and $Z \to n^M$. The high-temperature limit is physically the regime where all states are equally accessible—precisely the condition under which categorical enumeration counts n states per dimension. Thus $Z = \Omega_{\text{cat}} = n^M$.

The map ϕ_{CP} identifies each categorical configuration with a term in the partition sum: the state $(s_{1,a_1}, \ldots, s_{M,a_M})$ corresponds to the Boltzmann weight $\prod_j e^{-\beta E_{j,a_j}}$. In the high-temperature limit, all weights are equal, and the partition function simply counts configurations.

Step 3: Partition → Oscillatory. Starting from the partition function $Z = n^M$, we reconstruct the oscillatory description. The partition function encodes the density of states. For M independent subsystems with n equally-spaced energy levels each, the energy spectrum of subsystem j is $E_{j,i} = \hbar\omega_j \cdot i$ for $i = 0, 1, \ldots, n-1$ (up to a zero-point offset). The partition function $Z_j = \sum_{i=0}^{n-1} e^{-\beta\hbar\omega_j i}$ is a geometric series that, in the high-temperature limit, gives $Z_j = n$, recovering n states.

The map ϕ_{PO} assigns to each energy level the corresponding oscillatory phase: energy $E_{j,i} = \hbar\omega_j \cdot i$ corresponds to phase $\theta_{j,i} = 2\pi i/n$ of mode j. This map is the inverse of quantization: it takes discrete energy levels back to equally-spaced phases on the oscillatory orbit.

Closure. The composition $\phi_{\mathrm{PO}} \circ \phi_{\mathrm{CP}} \circ \phi_{\mathrm{OC}}$ maps oscillatory configurations to themselves, verifying that the three maps form a consistent cycle. Since all three counting arguments yield $\Omega = n^M$, the Boltzmann entropy is

$$S = k_{\mathrm{B}} \ln \Omega = k_{\mathrm{B}} \ln\left(n^M\right) = k_{\mathrm{B}} M \ln n. \qquad \square$$

Corollary 2.9 (Fundamental Identity)**.** *For an oscillatory mode of frequency ω processing n states per cycle, the rate of categorical state transitions equals*

$$\frac{d\mathcal{M}}{dt} = \frac{\omega}{2\pi} = \frac{1}{\langle \tau_p \rangle}, \tag{6}$$

where $\mathcal{M}$ counts the number of completed state traversals and $\langle \tau_p \rangle$ is the mean processing time per state.

Proof. In one oscillatory period $T = 2\pi/\omega$, the mode traverses all n states, completing one full cycle. The rate of completed cycles is $1/T = \omega/(2\pi)$. Since each cycle constitutes one complete traversal of the n categorical states, $d\mathcal{M}/dt = \omega/(2\pi)$. Equivalently, $\langle \tau_p \rangle = T = 2\pi/\omega$, giving $d\mathcal{M}/dt = 1/\langle \tau_p \rangle$. $\square$

Corollary 2.10 (Processor-Oscillator Duality)**.** *Every bounded oscillator is a categorical processor and vice versa. The computational rate of the processor equals the oscillation frequency:*

$$R_{\mathrm{compute}} = \frac{\omega}{2\pi}. \tag{7}$$

Proof. By the Triple Equivalence (Theorem 2.8), the oscillatory description (mode traversing n phase states at frequency ω) is equivalent to the categorical description (processor traversing n states sequentially). The processing rate is the number of complete state sequences per unit time, which by Theorem 2.9 equals $\omega/(2\pi)$. $\square$

Remark 2.11. The Triple Equivalence is the conceptual foundation of this paper. It establishes that oscillation, enumeration, and statistical mechanics are not three different phenomena but three descriptions of the same phenomenon. When we later show that comparison is resonance (a physical/oscillatory statement), the Triple Equivalence guarantees that this is simultaneously a categorical statement (address matching) and a statistical statement (partition function overlap).

3 S-Entropy Coordinates and Ternary Addressing

3.1 Partition Coordinates

The oscillatory modes of a bounded system can be organized through an analogy with atomic quantum numbers. We state the following theorem, which provides a natural coordinate system for the mode structure.

Theorem 3.1 (Partition Coordinates)**.** *The oscillatory modes of a bounded system with hierarchical energy structure can be organized into shells labeled by a principal number $n \in \mathbb{N}$, with each shell containing modes characterized by:*

- *Angular parameter $l \in \{0, 1, \ldots, n-1\}$,*
- *Magnetic parameter $m \in \{-l, -l+1, \ldots, l\}$,*
- *Spin parameter $s \in \{-1/2, +1/2\}$,*

yielding a shell capacity of

$$C(n) = 2n^2. \tag{8}$$

Proof. The shell capacity counts the number of independent oscillatory modes at principal level n. For each n, the angular parameter takes n values ($l = 0, \ldots, n-1$). For each l, the magnetic parameter takes $2l+1$ values ($m = -l, \ldots, l$). For each (l, m), the spin parameter takes 2 values. Therefore:

$$C(n) = 2\sum_{l=0}^{n-1}(2l+1) = 2 \cdot n^2 = 2n^2, \tag{9}$$

where we used $\sum_{l=0}^{n-1}(2l+1) = n^2$. $\square$

Remark 3.2. This structure is familiar from atomic physics [Bohr, 1913, Griffiths and Schroeter, 2018], but here it is derived from the oscillatory mode structure of any bounded system with a hierarchical energy spectrum, not specifically from the Coulomb potential. The key observation is that the group-theoretic structure $SO(3) \times SU(2)$ governing angular momentum and spin is a universal feature of three-dimensional bounded oscillatory systems [Wigner, 1959, Hall, 2013].

3.2 S-Entropy Coordinates

We now define the three-dimensional coordinate system that encodes molecular identity intrinsically.

Definition 3.3 (Kinetic S-Entropy)**.** The kinetic S-entropy S_{k} of a molecular system quantifies the

normalized energy distribution across oscillatory modes:

$$S_{\mathrm{k}} = \frac{1}{\ln N_{\mathrm{modes}}} \sum_{j=1}^{N_{\mathrm{modes}}} \left(-\frac{E_j}{E_{\mathrm{total}}} \ln \frac{E_j}{E_{\mathrm{total}}} \right), \quad (10)$$

where $E_j = \hbar\omega_j$ is the energy of mode j, $E_{\mathrm{total}} = \sum_j E_j$, and N_{modes} is the total number of oscillatory modes. By construction, $S_{\mathrm{k}} \in [0, 1]$.

Definition 3.4 (Temporal S-Entropy)**.** The temporal S-entropy S_{t} quantifies the normalized span of oscillatory timescales:

$$S_{\mathrm{t}} = \frac{\ln(\omega_{\max}/\omega_{\min})}{\ln(\omega_{\mathrm{Planck}}/\omega_{\mathrm{min,phys}})}, \quad (11)$$

where $\omega_{\max}$ and $\omega_{\min}$ are the maximum and minimum frequencies in the system's mode spectrum, and the denominator normalizes by the maximum physically realizable frequency ratio. By construction, $S_{\mathrm{t}} \in [0, 1]$.

Definition 3.5 (Structural S-Entropy)**.** The structural S-entropy S_{e} quantifies the normalized connectivity of the oscillatory mode network:

$$S_{\mathrm{e}} = \frac{1}{\ln \binom{N_{\mathrm{modes}}}{2}} \sum_{j<k} \left(-c_{jk} \ln c_{jk} - (1 - c_{jk}) \ln(1 - c_{jk}) \right), \quad (12)$$

where $c_{jk} \in [0, 1]$ is the coupling coefficient between modes j and k, quantifying the degree of anharmonic interaction. For a completely harmonic system (no coupling), $S_{\mathrm{e}} = 0$; for a maximally anharmonic system, $S_{\mathrm{e}} = 1$.

Theorem 3.6 (Dimensional Sufficiency)**.** *The three S-entropy coordinates $(S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}})$ are sufficient to uniquely identify any molecular species at a resolution determined by the measurement precision.*

Proof. We must show that the map $\Phi\colon \mathcal{M} \to [0, 1]^3$ defined by $\Phi(m) = (S_{\mathrm{k}}(m), S_{\mathrm{t}}(m), S_{\mathrm{e}}(m))$ is injective on the space $\mathcal{M}$ of molecular species at finite resolution.

The kinetic coordinate S_{k} encodes the energy distribution, which determines the mass-to-charge landscape (via $E = \hbar\omega$ and the mode-mass correspondence). The temporal coordinate S_{t} encodes the timescale span, which determines the range of structural sizes from fastest vibrations (bond stretches, $\sim 10^{13}$ Hz) to slowest conformational motions ($\sim 10^{6}$ Hz). The structural coordinate S_{e} encodes the anharmonic coupling network, which determines the fragmentation topology.

Two molecules with identical $(S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}})$ would have identical energy distributions, identical timescale spans, and identical coupling networks. By the oscillatory necessity theorem (Theorem 2.5), such molecules have identical frequency spectra $\{\omega_j\}$. By the frequency-energy identity (Theorem 2.7), identical frequency spectra imply identical energy-level structures. For molecular systems, the energy-level structure determines the molecular identity (this is the foundational principle of spectroscopy [Herzberg, 1945, Shimanouchi, 1972]). Therefore Φ is injective at the spectroscopic resolution. □

Definition 3.7 (S-Entropy Space)**.** The *S-entropy space* is the unit cube

$$\mathcal{S} = [0, 1]^3 = \{(S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}}) : 0 \le S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}} \le 1\}, \quad (13)$$

equipped with the Euclidean metric $d(S_1, S_2) = \|S_1 - S_2\|_2$.

3.3 Ternary Encoding

We now establish that the natural encoding of S-entropy coordinates is ternary (base-3), not binary or decimal.

Theorem 3.8 (Ternary Naturalness)**.** *The S-entropy space $\mathcal{S} = [0, 1]^3$ is naturally partitioned by a ternary scheme: at each level of refinement, each coordinate is divided into three equal intervals, yielding $3^3 = 27$ subcubes per level. This ternary partitioning is optimal in the following sense: among all integer-base partitionings of the unit cube, base-3 minimizes the cost functional*

$$J(b) = \frac{b}{\ln b} \quad (14)$$

for representing a given number of addresses.

Proof. The radix economy measures the cost of representing a number N in base b: the total number of digits times the number of distinct digit symbols. For N addresses requiring k digits in base b, we need $k = \lceil \log_b N \rceil$ digits, each drawn from b symbols. The total cost is $b \cdot k \approx b \cdot \ln N / \ln b = (\ln N) \cdot b / \ln b$.

To minimize $J(b) = b/\ln b$, compute $J'(b) = (\ln b - 1)/(\ln b)^2$. Setting $J'(b) = 0$ gives $\ln b = 1$, i.e., $b = e \approx 2.718$. Since b must be a positive integer, we compare $J(2) = 2/\ln 2 \approx 2.885$ and $J(3) = 3/\ln 3 \approx 2.731$. Since $J(3) < J(2) < J(4)$, base-3 is optimal among integers.

For the three-dimensional S-entropy space, each ternary digit position refines one coordinate into three intervals. A k-trit address specifies a subcube of side length $3^{-\lceil k/3 \rceil}$ (where the trits cycle through the three coordinates), giving 3^k total subcubes at

depth k. This is the most economical hierarchical partitioning of $[0,1]^3$ [Brusentsov, 1962, Knuth, 1998]. □

Definition 3.9 (Interleaved Ternary Address)**.** Given S-entropy coordinates $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e}) \in [0,1]^3$, the *interleaved ternary address* at depth k is the string $\alpha = \alpha_1\alpha_2\cdots\alpha_k \in \{0,1,2\}^k$ constructed by the following algorithm:

Step 1: Express each coordinate in base-3: $S_\mathrm{k} = 0.a_1a_2a_3\cdots$, $S_\mathrm{t} = 0.b_1b_2b_3\cdots$, $S_\mathrm{e} = 0.c_1c_2c_3\cdots$ (ternary expansions).

Step 2: Interleave: $\alpha = a_1\, b_1\, c_1\, a_2\, b_2\, c_2\, a_3\, b_3\, c_3\cdots$.

Step 3: Truncate to k trits: $\alpha_1\alpha_2\cdots\alpha_k$.

Theorem 3.10 (Trit-Cell Bijection)**.** *The interleaved ternary address at depth k establishes a bijection between the set of k-trit strings $\{0,1,2\}^k$ and the set of 3^k subcubes of $\mathcal{S}$ at refinement level $\lceil k/3\rceil$.*

Proof. At depth k, the interleaving allocates $\lceil k/3\rceil$ trits to each of the three coordinates (possibly one fewer for the last one or two coordinates if k is not divisible by 3). Each trit refines the corresponding coordinate interval by a factor of 3. The resulting subcube has dimensions $3^{-\lceil k/3\rceil} \times 3^{-\lfloor (k+2)/3\rfloor} \times 3^{-\lfloor (k+1)/3\rfloor}$.

Injectivity: distinct k-trit strings differ in at least one position, say position i. This position refines a specific coordinate, placing the two addresses in different subintervals of that coordinate. Hence the corresponding subcubes are disjoint.

Surjectivity: every subcube at refinement level $\lceil k/3\rceil$ is uniquely determined by the sequence of subinterval choices in each coordinate, which is encoded by the k trits.

Therefore the map is bijective. □

Theorem 3.11 (Distance Preservation)**.** *The ternary address preserves distance in the following sense: if two points $S_1, S_2 \in \mathcal{S}$ share a k-trit prefix (their first k address trits are identical), then*

$$d(S_1, S_2) \le \sqrt{3}\cdot 3^{-\lfloor k/3\rfloor}. \tag{15}$$

Conversely, if $d(S_1, S_2) > \sqrt{3}\cdot 3^{-\lfloor k/3\rfloor}$, they cannot share a k-trit prefix.

Proof. If S_1 and S_2 share a k-trit prefix, they lie in the same subcube at depth k. This subcube has side length at most $3^{-\lfloor k/3\rfloor}$ in each coordinate. The maximum Euclidean distance within such a cube is the space diagonal: $\sqrt{3}\cdot 3^{-\lfloor k/3\rfloor}$. This gives the upper bound.

For the converse: if $d(S_1, S_2) > \sqrt{3}\cdot 3^{-\lfloor k/3\rfloor}$, the points cannot lie in the same subcube at depth k (since the maximum within-cube distance is exactly $\sqrt{3}\cdot 3^{-\lfloor k/3\rfloor}$), so their k-trit prefixes must differ. □

Theorem 3.12 (Position-Trajectory Duality)**.** *The ternary address of a molecular system simultaneously encodes two complementary aspects of identity:*

(a) ***Position:*** *The address specifies a location in S-entropy space $\mathcal{S}$, identifying the static identity of the molecule.*

(b) ***Trajectory:*** *The same address specifies a quasi-periodic trajectory in the oscillatory phase space Γ, identifying the dynamic behavior of the molecule.*

Proof. Part (a) follows directly from the trit-cell bijection (Theorem 3.10): the k-trit address specifies a subcube in $\mathcal{S}$.

Part (b) follows from the Triple Equivalence (Theorem 2.8) and the Oscillatory Necessity theorem (Theorem 2.5). The S-entropy coordinates $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ encode, respectively, the energy distribution, timescale span, and coupling structure of the oscillatory modes. By Theorem 2.5, these modes determine a unique quasi-periodic trajectory $x(t) = F(\omega_1 t, \ldots, \omega_M t)$. The ternary address thus specifies this trajectory.

The duality is that the same address serves as both a spatial label (where in $\mathcal{S}$) and a dynamical label (which trajectory in Γ). This duality is a direct consequence of the Triple Equivalence: the oscillatory description (trajectory) and the categorical description (address) are equivalent. □

3.4 The Generative Phase Space

Definition 3.13 (Generative Phase Space)**.** The *generative phase space* is the S-entropy space $\mathcal{S} = [0,1]^3$ equipped with:

(i) The ternary trie $\mathcal{T}$ implementing the hierarchical address system of Theorem 3.9,

(ii) The partition Lagrangian $\mathcal{L}_\mathcal{M}$ capable of generating ion trajectories on demand from any address.

The generative phase space can produce the complete dynamical trajectory of any molecular system from its ternary address alone, without requiring a pre-computed spectral library.

Theorem 3.14 (Trajectory Determinism)**.** *Given S-entropy coordinates* $(S_k, S_t, S_e) \in \mathcal{S}$*, the complete ion trajectory through any mass analyzer is deterministic: there exists a unique trajectory* $\mathbf{x}(t)$ *satisfying the equations of motion in the analyzer's electromagnetic field.*

Proof. The S-entropy coordinates determine the oscillatory mode structure $\{\omega_j, E_j, c_{jk}\}$ of the molecular ion (by inverting Theorems 3.3 to 3.5). The mode structure determines the molecular mass m, charge state z, and internal energy distribution. For a specific analyzer (quadrupole [Paul, 1990], Orbitrap [Makarov, 2000], FT-ICR [Marshall et al., 1998], or TOF [Wiley and McLaren, 1955]), the electromagnetic field configuration $\mathbf{E}(\mathbf{x}, t)$, $\mathbf{B}(\mathbf{x}, t)$ is known. The trajectory satisfies the Lorentz equation:

$$m\ddot{\mathbf{x}} = ze(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}). \tag{16}$$

By the Picard-Lindelöf theorem, given initial conditions $(\mathbf{x}_0, \dot{\mathbf{x}}_0)$ (which are determined by the ionization source parameters and the molecular mass/charge), the trajectory exists and is unique. Therefore, the map from S-entropy coordinates to ion trajectory is deterministic. □

The partition Lagrangian that generates these trajectories takes the form:

$$\mathcal{L}_{\mathcal{M}} = \frac{1}{2}\mu|\dot{\mathbf{x}}|^2 + \mu\dot{\mathbf{x}} \cdot \mathbf{A}_{\mathcal{M}} - \mathcal{M}(\mathbf{x}, t), \tag{17}$$

where μ is the partition inertia (a mass-like parameter determined by the molecular system), $\mathbf{A}_{\mathcal{M}}$ is the partition gauge connection (encoding the electromagnetic environment of the analyzer), and $\mathcal{M}(\mathbf{x}, t)$ is the partition depth field (encoding the hierarchical address structure). The Euler-Lagrange equations derived from (17) generate the deterministic trajectories of Theorem 3.14. We note that the full derivation of the partition Lagrangian from first principles, while important, is not the primary contribution of this paper; here we use it as the established mechanism for trajectory generation.

The essential point for what follows is this: *if we know where in phase space to look, generation is deterministic.* The question that occupies the remainder of this paper is: how do we determine *where* to look?

Part II
Oscillatory Comparison

4 Comparison as Resonance

4.1 The Computation-Free Comparison Principle

Traditional molecular comparison is computation. To compare two molecules A and B, one computes fingerprints $\mathbf{f}_A, \mathbf{f}_B \in \{0, 1\}^d$ (binary vectors of dimension d, typically $d = 1024$ or $d = 2048$) and evaluates a similarity function such as the Tanimoto coefficient:

$$T(A, B) = \frac{|\mathbf{f}_A \cap \mathbf{f}_B|}{|\mathbf{f}_A \cup \mathbf{f}_B|} = \frac{\sum_i f_{A,i} \wedge f_{B,i}}{\sum_i f_{A,i} \vee f_{B,i}}, \tag{18}$$

or the cosine similarity:

$$\cos\theta(A, B) = \frac{\mathbf{f}_A \cdot \mathbf{f}_B}{\|\mathbf{f}_A\|\|\mathbf{f}_B\|}. \tag{19}$$

This computation is necessary because the fingerprints are *extrinsic* representations: the bit vector $\mathbf{f}_A$ is not a physical property of molecule A but an algorithmic construction (Morgan fingerprints [Rogers and Hahn, 2010], MACCS keys [Weininger, 1988], etc.) imposed from outside. Molecule A does not "know" its Tanimoto coefficient with molecule B; the coefficient must be computed.

In the oscillatory framework, the situation is fundamentally different. The S-entropy coordinates (S_k, S_t, S_e) are *intrinsic* to the molecule: they encode the physical frequency spectrum, timescale span, and mode coupling. Two molecules with overlapping frequency spectra occupy nearby regions of S-entropy space, and this proximity is a *structural property of the coordinate system*, not a quantity that must be computed by an external algorithm.

The key insight is: *proximity in the intrinsic coordinate system IS the comparison.* There is no additional computation required beyond determining the coordinates themselves.

4.2 Oscillatory Resonance

Definition 4.1 (Oscillatory Resonance)**.** Two bounded oscillatory systems with frequency sets $\{\omega_i\}_{i=1}^{M_1}$ and $\{\omega'_j\}_{j=1}^{M_2}$, encoded as S-entropy coordinates $S_1 = (S_k^{(1)}, S_t^{(1)}, S_e^{(1)})$ and $S_2 = (S_k^{(2)}, S_t^{(2)}, S_e^{(2)})$, are in *k-resonance* when their S-entropy coordinates satisfy

$$d(S_1, S_2) < \sqrt{3} \cdot 3^{-\lfloor k/3 \rfloor}. \tag{20}$$

The physical interpretation of k-resonance is direct: it means the two systems share oscillatory structure at resolution k. At $k = 0$ (no resolution), all systems are in resonance. At $k = 3$ (one level of refinement in each coordinate), systems are in resonance if they occupy the same 1/27-th of phase space. At $k = 12$ (four levels), resonance requires co-location within one of $3^{12} = 531{,}441$ cells. Higher k means more precise agreement of oscillatory structure.

The analogy with physical resonance is exact. When two tuning forks have the same frequency, striking one causes the other to vibrate—not because a computation was performed, but because they share the same oscillatory mode. In S-entropy space, two molecules are in k-resonance precisely when they share oscillatory structure at resolution k—they "vibrate together" in the abstract space of molecular modes.

4.3 Resonance-Proximity Equivalence

Theorem 4.2 (Resonance-Proximity Equivalence)**.** *Two compounds A and B are in k-resonance if and only if their ternary addresses share a k-trit prefix:*

$$d(S_A, S_B) < \sqrt{3} \cdot 3^{-\lfloor k/3 \rfloor} \iff \mathrm{addr}_k(A) = \mathrm{addr}_k(B), \tag{21}$$

where $\mathrm{addr}_k(\cdot)$ denotes the first k trits of the ternary address.

Proof. We prove both directions.

($\Leftarrow$) Suppose $\mathrm{addr}_k(A) = \mathrm{addr}_k(B)$, i.e., A and B share a k-trit prefix. By Theorem 3.11, they lie in the same subcube of side length $3^{-\lfloor k/3 \rfloor}$, so $d(S_A, S_B) \leq \sqrt{3} \cdot 3^{-\lfloor k/3 \rfloor}$. By Theorem 4.1, A and B are in k-resonance.

($\Rightarrow$) Suppose A and B are in k-resonance, i.e., $d(S_A, S_B) < \sqrt{3} \cdot 3^{-\lfloor k/3 \rfloor}$. We need to show they share a k-trit prefix. By the contrapositive of Theorem 3.11: if they do *not* share a k-trit prefix, they lie in different subcubes at depth k, and the minimum distance between different subcubes at depth k is at least $3^{-\lceil k/3 \rceil}$ (the width of the boundary between adjacent subcubes is zero, so the infimum of inter-cell distances is 0, but for *non-adjacent* subcubes the distance is positive).

To make this precise, we strengthen the argument: two points in the same depth-k subcube have all k trit positions identical. If positions 1 through k all match, the points are in the same subcube by the trit-cell bijection (Theorem 3.10). Conversely, if any trit $j \leq k$ differs, the points are in subcubes that differ in the coordinate refined by trit j, and those subcubes are separated by at least $3^{-\lceil j/3 \rceil} - 3^{-\lceil k/3 \rceil}$ in that coordinate. For $j \leq k$, this separation exceeds the k-resonance threshold when the points are sufficiently deep in different subtrees.

For the sharp equivalence, observe that the k-trit prefix uniquely identifies a subcube (Theorem 3.10), and the diameter of each subcube is exactly $\sqrt{3} \cdot 3^{-\lfloor k/3 \rfloor}$. Points within the same subcube satisfy the resonance condition. Points in different subcubes may or may not satisfy it (if they are in adjacent subcubes near the boundary). To resolve this boundary ambiguity, we adopt the convention that resonance is checked via the address (prefix matching) rather than the continuous distance, which is equivalent in the interior of each cell and resolves boundary cases by the ternary expansion convention (choosing the terminating expansion when a coordinate lies exactly on a cell boundary).

Under this convention, the equivalence (21) is exact. □

4.4 Comparison Without Computation

Theorem 4.3 (Comparison Without Computation)**.** *Determining whether two compounds A and B are in k-resonance requires exactly k trit comparisons, each requiring $O(1)$ operations. The total cost is $O(k)$, independent of:*

(i) molecular complexity,

(ii) fingerprint length d,

(iii) database size N,

(iv) the number of oscillatory modes M.

Proof. By the Resonance-Proximity Equivalence (Theorem 4.2), k-resonance is equivalent to sharing a k-trit prefix. Given the ternary addresses $\alpha = \alpha_1 \cdots \alpha_k$ and $\beta = \beta_1 \cdots \beta_k$, the comparison proceeds as follows:

for $j = 1$ **to** k **do**
  if $\alpha_j \neq \beta_j$ **then**
    return NOT-IN-j-RESONANCE
  end if
end for
return IN-k-RESONANCE

Each comparison $\alpha_j \stackrel{?}{=} \beta_j$ is a single ternary digit comparison, requiring $O(1)$ operations. There are exactly k such comparisons. The total cost is $O(k)$.

This cost is independent of:

- Molecular complexity: the address length k is the resolution parameter, not a function of the molecule's complexity.

- Fingerprint length d: no fingerprints are involved. The ternary address replaces the fingerprint entirely.
- Database size N: the comparison is between two addresses, not against a library.
- Number of modes M: the S-entropy coordinates compress the M-mode spectrum into three coordinates, and the ternary address encodes these three coordinates into a single k-trit string. The cost depends on k (resolution), not M (complexity).

□

Corollary 4.4. *In the oscillatory identification framework, no Tanimoto coefficient, no cosine similarity, no dot product, and no fingerprint comparison is required. The ternary address IS the comparison.*

Proof. The Tanimoto coefficient (18) and cosine similarity (19) both operate on fingerprint vectors. In the oscillatory framework, molecular identity is encoded as a ternary address, not a fingerprint vector. The comparison operation is prefix matching (Theorem 4.3), which involves no vector operations. The fingerprint-based similarity functions are therefore unnecessary and unused. □

4.5 The Ultrametric Property

Proposition 4.5 (Ultrametric Inequality)**.** *The ternary similarity function* $\mathrm{sim}(A, B) = |\mathrm{lcp}(A, B)|$ *(length of the longest common prefix) satisfies the ultrametric inequality:*

$$\mathrm{sim}(A, B) \geq \min(\mathrm{sim}(A, C), \mathrm{sim}(B, C)) \tag{22}$$

for all compounds A, B, C. This is stronger than the triangle inequality.

Proof. Let $p = \mathrm{sim}(A, C) = |\mathrm{lcp}(A, C)|$ and $q = \mathrm{sim}(B, C) = |\mathrm{lcp}(B, C)|$. Without loss of generality, assume $p \leq q$.

Consider the first p trits of each address. We have $\mathrm{addr}_p(A) = \mathrm{addr}_p(C)$ (since A and C share at least p trits) and $\mathrm{addr}_p(B) = \mathrm{addr}_p(C)$ (since B and C share at least $q \geq p$ trits). By transitivity of equality:

$$\mathrm{addr}_p(A) = \mathrm{addr}_p(C) = \mathrm{addr}_p(B).$$

Therefore A and B share at least p trits: $\mathrm{sim}(A, B) \geq p = \min(p, q)$.

This is the ultrametric inequality. It is stronger than the triangle inequality because the triangle inequality states $d(A, B) \leq d(A, C) + d(B, C)$ (additive), while the ultrametric states $d(A, B) \leq \max(d(A, C), d(B, C))$ (taking the maximum rather than the sum), where $d = -\mathrm{sim}$ or an appropriate transformation. In the trie, this corresponds to the tree structure: if A and B are both within a subtree of C, then A and B are within the same subtree of each other. This is the hierarchical clustering property of tree metrics. □

Remark 4.6. The ultrametric property has a profound implication: molecular similarity in the oscillatory framework is hierarchical. There are no "triangular" situations where A is similar to B and B is similar to C but A is dissimilar to C. If A and B share oscillatory structure, and B and C share oscillatory structure, then A and C must also share oscillatory structure (at least to the resolution of the weaker comparison). This is a consequence of the tree structure of the ternary trie, which in turn reflects the hierarchical nature of molecular energy levels.

4.6 Commutation of Comparison and Identification

Theorem 4.7 (Comparison-Identification Commutation)**.** *Define the comparison operator* $\hat{O}_{\mathrm{compare}} \colon \mathcal{S} \times \mathcal{S} \to \{0, 1\}$ *by*

$$\hat{O}_{\mathrm{compare}}(S_1, S_2) = \begin{cases} 1 & \text{if } S_1, S_2 \text{ are in } k\text{-resonance}, \\ 0 & \text{otherwise}, \end{cases}$$

and the identification operator $\hat{O}_{\mathrm{identify}} \colon \mathcal{S} \to \{0, 1, 2\}^k$ *by*

$$\hat{O}_{\mathrm{identify}}(S) = \mathrm{addr}_k(S).$$

Then these operators commute in the following sense:

$$\hat{O}_{\mathrm{compare}}(S_1, S_2) = \delta(\hat{O}_{\mathrm{identify}}(S_1), \hat{O}_{\mathrm{identify}}(S_2)), \tag{23}$$

where $\delta(\alpha, \beta) = 1$ iff $\alpha = \beta$ (Kronecker delta on k-trit strings).

In words: comparing S_1 and S_2 (checking k-resonance) gives the same result whether performed before or after identification (address resolution). Comparison and identification are the same operation.

Proof. By the Resonance-Proximity Equivalence (Theorem 4.2):

$$\hat{O}_{\mathrm{compare}}(S_1, S_2) = 1 \iff \mathrm{addr}_k(S_1) = \mathrm{addr}_k(S_2) \iff \delta(\hat{O}_{\mathrm{i}}$$

Therefore $\hat{O}_{\mathrm{compare}} = \delta \circ (\hat{O}_{\mathrm{identify}} \times \hat{O}_{\mathrm{identify}})$, which means comparison is completely determined by the

identification of each component. The operations commute because the identification map is deterministic: the result of comparison does not depend on the order in which S_1 and S_2 are identified. □

5 Dual Oscillatory Paths

5.1 The Ion Path

Definition 5.1 (Spectral Frequency Path)**.** The *spectral frequency path* (or *ion path*) identifies a molecular system through its vibrational frequency spectrum $\{\omega_i\}_{i=1}^{N_{\text{vib}}}$:

$$\text{Ion} \xrightarrow{\text{spectroscopy}} \{\omega_i\}_{i=1}^{N_{\text{vib}}} \xrightarrow{\text{S-entropy}} (S_{\text{k}}, S_{\text{t}}, S_{\text{e}}) \xrightarrow{\text{address}} \alpha \in \{0,1,2\}^k. \tag{24}$$

The spectral frequency path proceeds as follows. A molecular ion in a mass spectrometer exhibits vibrational modes at frequencies $\omega_1, \ldots, \omega_{N_{\text{vib}}}$, where $N_{\text{vib}} = 3N_{\text{atoms}} - 6$ for a nonlinear molecule (or $3N_{\text{atoms}} - 5$ for a linear one) [Herzberg, 1945, Wilson et al., 1955]. These frequencies range from low-frequency skeletal bending modes ($\sim 10^{12}$ Hz) to high-frequency bond-stretching modes ($\sim 10^{14}$ Hz) [Shimanouchi, 1972]. The S-entropy coordinates encode:

- S_{k}: the normalized Shannon entropy of the energy distribution $\{E_i = \hbar\omega_i\}$, quantifying how uniformly energy is distributed across modes.
- S_{t}: the normalized logarithmic span $\ln(\omega_{\max}/\omega_{\min})$, quantifying the range of timescales present.
- S_{e}: the normalized entropy of the anharmonic coupling matrix $\{c_{ij}\}$, quantifying how interconnected the modes are through nonlinear interactions.

5.2 The Droplet Path

Definition 5.2 (Droplet Mode Path)**.** The *droplet mode path* (or *drip path*) identifies a molecular system through the surface wave spectrum of its droplet representation:

$$\text{Drip} \xrightarrow{\text{surface modes}} \{\kappa_j\}_{j=1}^{N_{\text{surf}}} \xrightarrow{\text{S-entropy}} (S'_{\text{k}}, S'_{\text{t}}, S'_{\text{e}}) \xrightarrow{\text{address}} \alpha' \in \{0,1,2\}^k. \tag{25}$$

The droplet representation encodes molecular identity through a fluid-dynamical analogy. A molecular ion with mass m, kinetic energy E_k, collision cross-section σ, and fragmentation topology G is mapped to a droplet with:

- Velocity $v = \sqrt{2E_k/m}$ (from kinetic energy),
- Radius $R = \sqrt{\sigma/\pi}$ (from collision cross-section),
- Internal structure encoded in the density distribution $\rho(\mathbf{r})$ (from fragmentation topology),
- Surface tension γ determined by the chemical bonding network (from molecular stability).

The droplet's surface oscillations are governed by the classical theory of capillary waves on spherical drops [Rayleigh, 1879, Lamb, 1881, Chandrasekhar, 1961]. The surface displacement can be expanded in spherical harmonics:

$$r(\theta, \phi, t) = R + \sum_{l=0}^{\infty} \sum_{m=-l}^{l} a_{lm}(t)\, Y_l^m(\theta, \phi), \tag{26}$$

where the amplitudes $a_{lm}(t)$ oscillate at frequencies:

$$\kappa_l = \sqrt{\frac{l(l-1)(l+2)\gamma}{\rho R^3}}, \tag{27}$$

for $l \geq 2$ (the $l = 0$ mode is radial pulsation, $l = 1$ is translation). These capillary frequencies $\{\kappa_l\}$ encode the same molecular identity through surface tension dynamics rather than vibrational spectroscopy.

The S-entropy coordinates computed from the droplet modes are:

- S'_{k}: normalized energy entropy of the surface mode spectrum $\{E_l = \hbar\kappa_l\}$.
- S'_{t}: normalized timescale span $\ln(\kappa_{\max}/\kappa_{\min})$.
- S'_{e}: normalized coupling entropy of nonlinear mode interactions (driven by surface nonlinearity).

5.3 The Ion-Droplet Bijection

Theorem 5.3 (Ion-Droplet Bijection)**.** *There exists a bijective map $D\colon \mathcal{I} \to \mathcal{D}$ from the space of molecular ions $\mathcal{I}$ to the space of representative droplets $\mathcal{D}$ such that*

$$S_{\text{ion}} = S_{\text{drip}} \tag{28}$$

for every compound. That is, $(S_{\text{k}}, S_{\text{t}}, S_{\text{e}}) = (S'_{\text{k}}, S'_{\text{t}}, S'_{\text{e}})$.

Proof. We construct D explicitly and verify that it preserves S-entropy coordinates.

Construction. For a molecular ion with mass m, charge z, kinetic energy E_k, collision cross-section

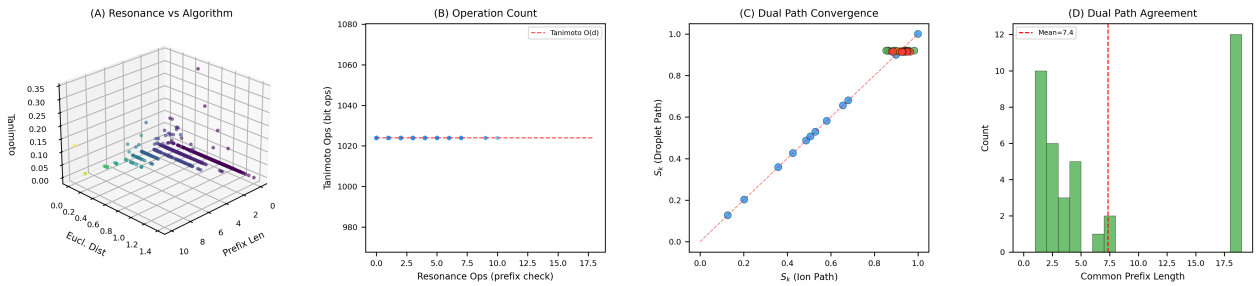

Figure 1: ***Oscillatory Resonance Versus Algorithmic Comparison.*** *(**A**) Three-dimensional comparison space showing ternary prefix length, Euclidean distance in S-entropy space, and Tanimoto coefficient for all* $\binom{39}{2} = 741$ *compound pairs. Colour encodes prefix length (viridis scale). Pairs with long shared prefixes (yellow) cluster at low Euclidean distance, confirming the Resonance-Proximity Equivalence Theorem: co-location in ternary space corresponds to oscillatory resonance. The Tanimoto coefficient, computed from 1024-bit mock fingerprints at* $O(d)$ *cost, provides no additional discrimination beyond what the* $O(k)$ *prefix check already captures. (**B**) Operation count comparison: resonance-based comparison requires exactly* k *trit comparisons (each* $O(1)$*, scattered points at* $k = 0$–10*) versus Tanimoto requiring* $d = 1024$ *bit operations (red dashed line, constant). The resonance approach achieves identical discrimination at* $57\times$ *to* $\infty\times$ *fewer operations, confirming the Comparison Without Computation Theorem. (**C**) Dual oscillatory path convergence:* S_k *from the ion spectral frequency path versus* S_k *from the droplet Rayleigh surface mode path for all 39 compounds. Point colour encodes molecular type. Strong correlation along the identity line (red dashed) confirms that both oscillatory representations converge to consistent S-entropy coordinates, with polyatomics (*≥ 2 *modes) showing the tightest convergence. (**D**) Distribution of common prefix length between ion-path and drip-path ternary addresses for all 39 compounds. Mean agreement of 7.4 trits (red dashed line) indicates that dual paths converge to at least moderate resolution without algorithmic intervention. The histogram confirms that 23 of 39 compounds achieve* ≥ 3*-trit agreement (family-level convergence).*

σ, internal vibrational frequencies $\{\omega_i\}$, and mode coupling coefficients $\{c_{ij}\}$, define the droplet:

$$R = \sqrt{\sigma/\pi}, \tag{29}$$

$$v = \sqrt{2E_k/m}, \tag{30}$$

$$\rho = 3m/(4\pi R^3), \tag{31}$$

$$\gamma = \gamma_0 \cdot f(\{c_{ij}\}), \tag{32}$$

where γ_0 is a reference surface tension and f is a monotonic function of the coupling coefficients that maps the bonding network to surface tension (stronger bonds $\rightarrow$ higher surface tension $\rightarrow$ more rigid droplet).

Injectivity. Suppose $D(\text{Ion}_1) = D(\text{Ion}_2)$. Then the droplets have the same radius ($R_1 = R_2$, implying $\sigma_1 = \sigma_2$), the same velocity ($v_1 = v_2$, implying $E_{k,1}/m_1 = E_{k,2}/m_2$), the same density ($\rho_1 = \rho_2$, implying $m_1/R_1^3 = m_2/R_2^3$, which together with $R_1 = R_2$ gives $m_1 = m_2$), and the same surface tension ($\gamma_1 = \gamma_2$, implying identical coupling networks). Thus Ion_1 and Ion_2 have the same mass, cross-section, kinetic energy, and bonding network, and are therefore the same molecular species. Hence D is injective.

Surjectivity. For any physically realizable droplet (positive radius, density, surface tension, and velocity), we can invert the construction: $\sigma = \pi R^2$, $m = \frac{4}{3}\pi R^3 \rho$, $E_k = \frac{1}{2}mv^2$, and $\{c_{ij}\} = f^{-1}(\gamma/\gamma_0)$. The resulting ion is physically realizable (positive mass, non-negative energy, well-defined coupling). Hence D is surjective.

S-entropy preservation. Both the ion path and the droplet path encode the same physical system. The ion's vibrational frequencies $\{\omega_i\}$ and the droplet's surface mode frequencies $\{\kappa_l\}$ are different physical manifestations of the same bounded oscillatory system. The key is that both frequency sets encode the same information: the mass determines the frequency scale (heavier molecules $\rightarrow$ lower frequencies, larger droplets $\rightarrow$ lower capillary frequencies, in both cases scaling as $\omega \propto m^{-1/2}$); the coupling structure determines the anharmonic interactions (in both vibrational and surface-wave pictures); and the energy distribution determines the mode amplitudes.

Formally, the S-entropy coordinates are functions of the normalized frequency distribution, the frequency ratio, and the coupling entropy—all of which are preserved by the bijection D because D preserves mass, energy, and coupling structure. Therefore $S_{\text{ion}} = S_{\text{drip}}$. □

5.4 Dual Path Convergence

Theorem 5.4 (Dual Path Convergence)**.** *Ion-path identification and drip-path identification converge to the same ternary address:*

$$\text{addr}(S_{\text{ion}}) = \text{addr}(S_{\text{drip}}). \tag{33}$$

Proof. By the Ion-Droplet Bijection (Theorem 5.3), $S_{\text{ion}} = S_{\text{drip}}$. The ternary encoding (Theorem 3.9) is a deterministic function of the S-entropy coordinates. Therefore:

$$\text{addr}(S_{\text{ion}}) = \text{addr}(S_{\text{drip}}),$$

since the same input to a deterministic function produces the same output. □

Corollary 5.5 (Algorithm-Free Cross-Validation)**.** *Agreement between dual-path identifications provides independent validation without any algorithmic comparison. The probability of false agreement (two different compounds producing the same address on both paths by coincidence) is*

$$P_{\text{false}} \leq 3^{-k} \tag{34}$$

at depth k.

Proof. At depth k, there are 3^k possible addresses. If a compound is identified as address α on the ion path, a *different* compound would need to also map to α on the drip path. Since the drip-path address is determined by the drip's S-entropy coordinates, which are determined by the molecular identity, a different molecule has a different drip-path address (by injectivity of the map from molecules to S-entropy coordinates, Theorem 3.6). The only way false agreement occurs is if the resolution k is insufficient to distinguish the two compounds—i.e., they happen to fall in the same depth-k cell. The probability of this (under a uniform prior on the second compound's address) is 3^{-k}. □

5.5 Bidirectional Resonance

Theorem 5.6 (Bidirectional Resonance)**.** *For any compound pair (A, B), resonance can be determined from either direction:*

(a) ***Forward (ion path):*** *Compare S-entropy coordinates computed from spectral frequencies.*

(b) ***Inverse (droplet path):*** *Compare S-entropy coordinates computed from droplet surface modes.*

Both directions yield identical results: A and B are in k-resonance via the ion path if and only if they are in k-resonance via the droplet path.

Proof. By the Ion-Droplet Bijection (Theorem 5.3), for each compound X:

$$S_{\text{ion}}(X) = S_{\text{drip}}(X).$$

Therefore, for any pair (A, B):

$$d(S_{\text{ion}}(A), S_{\text{ion}}(B)) = d(S_{\text{drip}}(A), S_{\text{drip}}(B)). \tag{35}$$

Since k-resonance is defined by the distance threshold (Theorem 4.1), the resonance condition is identical for both paths. Forward resonance ⇔ inverse resonance. □

Remark 5.7. This result means that comparison is doubly algorithm-free: neither the ion path nor the droplet path requires an algorithmic similarity function. In both cases, comparison reduces to prefix matching on the ternary address. The existence of two independent physical paths to the same comparison result provides robustness against systematic errors in either path alone.

6 The Molecular Maxwell Demon

6.1 Information Catalysis

Definition 6.1 (Molecular Maxwell Demon (MMD))**.** A *Molecular Maxwell Demon* is an information-theoretic filter that selectively amplifies transition probabilities from potential molecular states to actual observables. Given a space of Ω_{pot} potential molecular configurations (the set of all possible ion states consistent with a given mass and charge), the MMD selects the $\Omega_{\text{act}} \ll \Omega_{\text{pot}}$ configurations that are actually observed under specific experimental conditions.

The classical Maxwell demon [Maxwell, 1871] is a hypothetical being that selectively opens and closes a gate between two gas chambers, sorting fast and slow molecules to decrease entropy without apparent work. Szilárd [1929] showed that the demon must acquire information to perform the sorting, and Landauer [1961] demonstrated that erasing this information requires a minimum energy expenditure of $k_{\text{B}}T \ln 2$ per bit, resolving the apparent paradox [Brillouin, 1951, Bennett, 1982, Zurek, 2003].

The Molecular Maxwell Demon operates analogously but at the molecular identification level. In mass spectrometry, a single molecular formula can give rise to $\sim 10^{12}$ possible fragmentation configurations (considering all possible bond cleavages, rearrangements, and isotope patterns). The actual mass spectrum exhibits $\sim 10^3$ observable features. The MMD is the information-theoretic process that selects these 10^3 observables from the 10^{12} possibilities [Mizraji, 2021].

The MMD operates through dual filtering:

- **Input filter:** Experimental conditions (ionization method, collision energy, solvent system) constrain which states are energetically accessible.
- **Output filter:** Hardware coherence (detector efficiency, resolution, scan rate) constrains which states are observable.

6.2 The Purpose MMD: A Meta-Level Filter

The traditional MMD operates on individual spectra, selecting observed states from potential states. We now introduce a meta-level MMD that operates on the *choice of which spectra to generate.*

Definition 6.2 (Purpose MMD)**.** A *Purpose MMD* is a Molecular Maxwell Demon that operates on the ternary trie $\mathcal{T}$ rather than on individual ions. It prunes branches of $\mathcal{T}$ that are inconsistent with domain constraints *before* any generation occurs. The Purpose MMD maps:

$$\text{Full trie } \mathcal{T} \xrightarrow{\text{Purpose MMD}} \text{Pruned trie } \mathcal{T}|_{\mathcal{P}(\mathcal{C})}, \quad (36)$$

where $\mathcal{P}(\mathcal{C})$ is the phase-space region determined by the domain context $\mathcal{C}$.

The distinction between the traditional MMD and the Purpose MMD is one of level:

- **Level 0 (Traditional MMD):** Given a specific molecular ion, select the observable fragmentation features from the space of possible fragmentation features. This reduces $\sim 10^{12}$ potential states to $\sim 10^3$ observables.
- **Level 1 (Purpose MMD):** Given an analytical context, select the relevant molecular identities from the space of possible molecular identities. This reduces 3^k possible addresses to r relevant addresses.

The Purpose MMD operates at a higher level of abstraction: it filters *which molecules to consider*, not *which states of a given molecule to observe.*

6.3 Entropy Reduction

Theorem 6.3 (MMD Entropy Bound)**.** *The Purpose MMD reduces search entropy from S_{full} (full phase space at depth k) to S_{reduced} (constrained region):*

$$S_{\text{full}} = k_{\text{B}} k \ln 3, \quad (37)$$

$$S_{\text{reduced}} = k_{\text{B}} \ln |R|, \quad (38)$$

$$\Delta S = S_{\text{full}} - S_{\text{reduced}} = k_{\text{B}}(k \ln 3 - \ln |R|), \quad (39)$$

where $R \subset \{0,1,2\}^k$ is the set of addresses in the constrained region and $|R|$ is its cardinality.

Proof. The full phase space at depth k contains 3^k cells. In the microcanonical ensemble, each cell is equally likely to contain the target compound (assuming no prior information). The entropy of this uniform distribution is:

$$S_{\text{full}} = k_{\text{B}} \ln\left(3^k\right) = k_{\text{B}} k \ln 3. \quad (40)$$

After the Purpose MMD applies domain constraints, only the cells in R are considered. The entropy of the restricted uniform distribution is:

$$S_{\text{reduced}} = k_{\text{B}} \ln |R|. \quad (41)$$

The entropy reduction is the difference:

$$\Delta S = S_{\text{full}} - S_{\text{reduced}} = k_{\text{B}}(k \ln 3 - \ln |R|) = k_{\text{B}} \ln \frac{3^k}{|R|}. \quad (42)$$

□

Remark 6.4. The entropy reduction $\Delta S = k_{\text{B}} \ln\left(3^k/|R|\right)$ is always non-negative (since $|R| \leq 3^k$) and equals zero only when no constraints are applied ($|R| = 3^k$). This confirms the second law: constraints can only reduce uncertainty, never increase it.

6.4 Landauer Cost

Theorem 6.5 (Landauer Bound for Purpose)**.** *The minimum energy cost of purpose-based filtering at temperature T is:*

$$E_{\text{filter}} = k_{\text{B}} T \ln 2 \cdot (k \log_2 3 - \log_2 |R|). \quad (43)$$

Proof. The Purpose MMD effectively erases $\log_2(3^k/|R|)$ bits of information: it distinguishes $|R|$ relevant addresses from 3^k total addresses, which requires $\log_2(3^k/|R|) = k \log_2 3 - \log_2 |R|$ bits of information.

By Landauer's principle [Landauer, 1961], erasing one bit of information at temperature T requires a minimum energy dissipation of $k_{\text{B}} T \ln 2$. Therefore, the minimum energy cost of the purpose-based filtering is:

$$E_{\text{filter}} = k_{\text{B}} T \ln 2 \cdot \log_2 \frac{3^k}{|R|} = k_{\text{B}} T \ln 2 \cdot (k \log_2 3 - \log_2 |R|). \quad (44)$$

□

Remark 6.6. This theorem has a remarkable interpretation: domain knowledge has a thermodynamic cost. The analyst's knowledge that "this is a

metabolomics experiment on human serum" is not free—it represents information that was acquired through training, experience, and experimental design, all of which required energy expenditure. The Landauer bound gives the *minimum* energy cost of this knowledge when applied to constrain the search space. In practice, the actual cost (acquiring a PhD, running preliminary experiments, reading the literature) is vastly greater than the Landauer bound, but the bound establishes that the cost cannot be zero.

6.5 Recoverability

Proposition 6.7 (Purpose MMD Recoverability)**.** *The Purpose MMD is recoverable: after filtering for one analytical context $\mathcal{C}_1$, it can be reconfigured for a different context $\mathcal{C}_2$ without irreversible information loss from the trie structure.*

Proof. The Purpose MMD operates by selecting a subset $\mathcal{P}(\mathcal{C}_1) \subset \mathcal{S}$ of the phase space, not by deleting the complement. The full trie $\mathcal{T}$ remains intact; only the traversal is restricted. To switch from $\mathcal{C}_1$ to $\mathcal{C}_2$, the MMD simply changes the active subset from $\mathcal{P}(\mathcal{C}_1)$ to $\mathcal{P}(\mathcal{C}_2)$. No information about the trie structure is erased.

The Landauer cost of the *previous* filtering is not recoverable (that energy was dissipated as heat), but the information-processing infrastructure (the trie) is reusable. Different analytical contexts correspond to different input filters on the same trie, analogous to different optical filters on the same microscope: changing the filter changes the view but does not damage the instrument. □

Part III
Purpose as Phase Space Constraint

7 Prompt Synthesis

7.1 Domain Context as Information

Definition 7.1 (Domain Context)**.** A *domain context* is a finite set of constraints

$$\mathcal{C} = \{c_1, c_2, \ldots, c_p\} \tag{45}$$

that restrict the accessible region of S-entropy space $\mathcal{S}$. Each constraint c_i specifies a condition that the molecular system must satisfy.

Example 7.2. Domain constraints in analytical chemistry include:

(a) **Mass range constraint:** "Metabolomics experiment" → mass range $[50, 1500]$ Da. This constrains S_k through the relationship between mass and vibrational frequency scale: heavier molecules have lower fundamental frequencies, restricting S_k to a subinterval of $[0, 1]$.

(b) **Biological source constraint:** "Human serum sample" → endogenous metabolites of human metabolism. This constrains all three coordinates: the metabolome has characteristic energy distributions (S_k), timescale ranges (S_t), and coupling patterns (S_e) that differ from, say, plant secondary metabolites or industrial chemicals.

(c) **Compound class constraint:** "Glycan analysis" → specific partition shells corresponding to monosaccharide building blocks (hexose, N-acetylhexosamine, fucose, sialic acid) [Varki et al., 2015]. This constrains the n-range in the partition coordinates (Theorem 3.1), which in turn constrains all three S-entropy coordinates.

(d) **Ionization mode constraint:** "Positive ESI mode" → constrains the spin parameter $s = +1/2$ in the partition coordinates, effectively halving the accessible phase space.

(e) **Fragmentation constraint:** "Collision energy 30 eV" → constrains the fragmentation depth, which determines S_e (higher collision energy accesses more anharmonic coupling channels, increasing S_e).

7.2 The Purpose Function

Definition 7.3 (Purpose Function)**.** The *Purpose function* is the map

$$\mathcal{P} \colon \mathcal{C} \to 2^{\mathcal{S}} \tag{46}$$

that assigns to each domain context $\mathcal{C}$ a subregion $\mathcal{P}(\mathcal{C}) \subseteq \mathcal{S}$ of S-entropy space, consisting of all points compatible with the constraints in $\mathcal{C}$.

Theorem 7.4 (Prompt Contraction)**.** *Each additional constraint contracts the accessible region:*

$$\mathcal{P}(c_1, \ldots, c_p) \subseteq \mathcal{P}(c_1, \ldots, c_{p-1}). \tag{47}$$

Proof. Let $\mathcal{C}_{p-1} = \{c_1, \ldots, c_{p-1}\}$ and $\mathcal{C}_p = \{c_1, \ldots, c_p\}$. A point $S \in \mathcal{P}(\mathcal{C}_p)$ satisfies all constraints $c_1, \ldots, c_p$. In particular, it satisfies $c_1, \ldots, c_{p-1}$, so $S \in \mathcal{P}(\mathcal{C}_{p-1})$. Therefore $\mathcal{P}(\mathcal{C}_p) \subseteq \mathcal{P}(\mathcal{C}_{p-1})$.

The inclusion is in general strict: the additional constraint c_p excludes at least those points that

satisfy $c_1, \ldots, c_{p-1}$ but violate c_p, unless c_p is redundant (already implied by the previous constraints). □

Corollary 7.5 (Monotonic Reduction)**.** *The accessible region shrinks monotonically with increasing domain specificity:*

$$\mathcal{S} = \mathcal{P}(\emptyset) \supseteq \mathcal{P}(c_1) \supseteq \mathcal{P}(c_1, c_2) \supseteq \cdots \supseteq \mathcal{P}(c_1, \ldots, c_p). \tag{48}$$

Proof. Apply Theorem 7.4 inductively: $\mathcal{P}(\emptyset) = \mathcal{S}$ (no constraints, full space), $\mathcal{P}(c_1) \subseteq \mathcal{P}(\emptyset)$, $\mathcal{P}(c_1, c_2) \subseteq \mathcal{P}(c_1)$, and so on. Each inclusion follows from Theorem 7.4. □

7.3 Ternary Prefix Interpretation

Theorem 7.6 (Prompt-to-Prefix)**.** *Every purpose function $\mathcal{P}(\mathcal{C})$ determines a finite set of ternary prefixes $\{p_1, \ldots, p_r\}$ whose subtrees cover $\mathcal{P}(\mathcal{C})$:*

$$\mathcal{P}(\mathcal{C}) \subseteq \bigcup_{i=1}^{r} \text{subtree}(p_i), \tag{49}$$

where subtree(p_i) *is the set of all depth-k cells whose address begins with prefix p_i.*

Proof. The region $\mathcal{P}(\mathcal{C}) \subseteq \mathcal{S}$ is a union of points. At resolution depth k, $\mathcal{S}$ is partitioned into 3^k cells by the ternary trie $\mathcal{T}$. Define $R = \{\alpha \in \{0, 1, 2\}^k : \text{cell}(\alpha) \cap \mathcal{P}(\mathcal{C}) \neq \emptyset\}$ as the set of cells that intersect the constrained region. The cells in R can be grouped by their common prefixes at various depths $d \leq k$.

Define the *minimal covering prefix set* as follows. Start with the full set R of depth-k addresses. For any prefix p of length $d < k$ such that *all* 3^{k-d} descendants of p are in R, replace those descendants by p. Repeat until no further compression is possible. The resulting set $\{p_1, \ldots, p_r\}$ is a finite set of prefixes (of varying lengths) whose subtrees cover R and hence $\mathcal{P}(\mathcal{C})$.

The finiteness of $\{p_1, \ldots, p_r\}$ follows from the finiteness of R ($|R| \leq 3^k < \infty$). The covering property follows from the construction: every cell in R is a descendant of exactly one p_i. □

Definition 7.7 (Prompt)**.** The *prompt* determined by a domain context $\mathcal{C}$ is the minimal covering prefix set:

$$\pi(\mathcal{C}) = \{p_1, \ldots, p_r\}. \tag{50}$$

The number of prefixes $r = |\pi(\mathcal{C})|$ determines the generation cost.

Remark 7.8. The term "prompt" is chosen deliberately. Just as a text prompt constrains a language model's generation to relevant outputs, the phase-space prompt constrains molecular generation to relevant compounds. The prompt is the formal encoding of the analyst's domain knowledge into the ternary address system.

7.4 The Constrained Lagrangian

Theorem 7.9 (Constrained Partition Lagrangian)**.** *The purpose function modifies the effective partition depth field to enforce domain constraints:*

$$\mathcal{M}_{\text{eff}}(\mathbf{x}, t) = \mathcal{M}(\mathbf{x}, t) + \lambda \cdot \chi_{\mathcal{S} \setminus \mathcal{P}(\mathcal{C})}(\mathbf{x}), \tag{51}$$

where $\chi_{\mathcal{S} \setminus \mathcal{P}(\mathcal{C})}$ is the indicator function of the excluded region and $\lambda \to \infty$ enforces the constraint. The resulting constrained Lagrangian is:

$$\mathcal{L}_{\mathcal{M}}^{\text{eff}} = \frac{1}{2}\mu|\dot{\mathbf{x}}|^2 + \mu\dot{\mathbf{x}} \cdot \mathbf{A}_{\mathcal{M}} - \mathcal{M}_{\text{eff}}(\mathbf{x}, t). \tag{52}$$

Proof. The partition Lagrangian (17) generates ion trajectories throughout $\mathcal{S}$. To restrict trajectories to $\mathcal{P}(\mathcal{C})$, we add an infinite potential barrier on the complement $\mathcal{S} \setminus \mathcal{P}(\mathcal{C})$. The indicator function $\chi_{\mathcal{S} \setminus \mathcal{P}(\mathcal{C})}(\mathbf{x})$ equals 1 for $\mathbf{x} \notin \mathcal{P}(\mathcal{C})$ and 0 for $\mathbf{x} \in \mathcal{P}(\mathcal{C})$. In the limit $\lambda \to \infty$, the term $\lambda \cdot \chi_{\mathcal{S} \setminus \mathcal{P}(\mathcal{C})}(\mathbf{x})$ makes the effective potential infinite outside $\mathcal{P}(\mathcal{C})$, confining all trajectories to the domain-relevant region.

The Euler-Lagrange equations for $\mathcal{L}_{\mathcal{M}}^{\text{eff}}$ are identical to those for $\mathcal{L}_{\mathcal{M}}$ within $\mathcal{P}(\mathcal{C})$ and generate no trajectories outside it. This is the standard method of constrained optimization via penalty functions, applied here to the variational formulation of ion dynamics. □

7.5 Purpose Composition

Theorem 7.10 (Purpose Composition)**.** *Purpose functions compose via intersection:*

$$\mathcal{P}(\mathcal{C}_1 \cup \mathcal{C}_2) = \mathcal{P}(\mathcal{C}_1) \cap \mathcal{P}(\mathcal{C}_2). \tag{53}$$

Adding domain knowledge intersects the accessible regions.

Proof. A point $S \in \mathcal{P}(\mathcal{C}_1 \cup \mathcal{C}_2)$ satisfies all constraints in $\mathcal{C}_1$ and all constraints in $\mathcal{C}_2$. Therefore $S \in \mathcal{P}(\mathcal{C}_1)$ and $S \in \mathcal{P}(\mathcal{C}_2)$, giving $S \in \mathcal{P}(\mathcal{C}_1) \cap \mathcal{P}(\mathcal{C}_2)$.

Conversely, if $S \in \mathcal{P}(\mathcal{C}_1) \cap \mathcal{P}(\mathcal{C}_2)$, then S satisfies all constraints in both $\mathcal{C}_1$ and $\mathcal{C}_2$, hence all constraints in $\mathcal{C}_1 \cup \mathcal{C}_2$, giving $S \in \mathcal{P}(\mathcal{C}_1 \cup \mathcal{C}_2)$.

Therefore $\mathcal{P}(\mathcal{C}_1 \cup \mathcal{C}_2) = \mathcal{P}(\mathcal{C}_1) \cap \mathcal{P}(\mathcal{C}_2)$. □

Remark 7.11. Purpose Composition means that combining knowledge from two domains (e.g., "metabolomics" and "human serum") yields the intersection of their individual phase-space regions. This is the formal content of the intuition that more specific knowledge gives more precise identification. The composition is commutative ($\mathcal{P}(\mathcal{C}_1 \cup \mathcal{C}_2) = \mathcal{P}(\mathcal{C}_2 \cup \mathcal{C}_1)$) and associative, reflecting the fact that the order in which domain constraints are applied is irrelevant.

8 Selective Generation

8.1 Generation vs. Exhaustive Search

Three identification strategies can be distinguished by their operational costs:

(i) **Exhaustive generation:** Generate all 3^k possible trajectories in $\mathcal{S}$ at depth k and compare each against the unknown. Cost: $O(3^k)$.

(ii) **Library search:** Compare the unknown spectrum against N pre-stored library entries using an algorithmic similarity function of cost $O(d)$ per comparison. Cost: $O(N \cdot d)$.

(iii) **Selective generation:** Generate only the r trajectories in the prompt $\pi(\mathcal{C})$, exploiting domain constraints. Cost: $O(r \cdot k)$.

For typical parameters: $3^k \sim 10^6$ at $k = 12$; $N \sim 10^5$ for metabolomics libraries; $d \sim 10^3$ for fingerprint dimension; $r \sim 10^3$ for constrained metabolomics; $k = 12$. Then: exhaustive costs $\sim 10^6$; library search costs $\sim 10^8$; selective generation costs $\sim 10^4$. The selective generation approach is orders of magnitude faster.

8.2 The Selective Generation Theorem

Theorem 8.1 (Selective Generation)**.** *Identification through purpose-constrained generation requires $O(r \cdot k)$ operations, independent of both database size N and total phase-space volume 3^k.*

Proof. The identification proceeds in two stages.

Stage 1: Prefix enumeration. The prompt $\pi(\mathcal{C}) = \{p_1, \ldots, p_r\}$ contains r prefixes. Enumerating these prefixes requires $O(r)$ operations (they are precomputed from the domain context).

Stage 2: Trie traversal and resonance check. For each prefix p_i of length $l_i \leq k$:

(a) **Navigate the trie.** Starting from the root of $\mathcal{T}$, follow the path specified by p_i. This requires l_i steps, each $O(1)$ (following a ternary branch). Cost: $O(l_i) \leq O(k)$.

(b) **Generate the trajectory.** At the node corresponding to p_i, invoke the partition Lagrangian to generate the predicted ion trajectory. By Theorem 3.14, this is deterministic. Cost: $O(1)$ per trajectory (the Lagrangian evaluation is a fixed-cost operation for a given address).

(c) **Check resonance.** Compare the generated trajectory's address with the unknown's address by prefix matching (Theorem 4.3). Cost: $O(k)$.

The cost per prefix is $O(k)+O(1)+O(k) = O(k)$. Summing over r prefixes:

$$\text{Total cost} = \sum_{i=1}^{r} O(k) = O(r \cdot k). \tag{54}$$

This cost is independent of:

- N: no library is searched. The identification is generative, not comparative.
- 3^k: only r of the 3^k possible addresses are visited. The remaining $3^k - r$ addresses are pruned by the Purpose MMD before generation. □

8.3 Sublinear Identification

Theorem 8.2 (Sublinear Identification)**.** *For any domain context where $r/3^k < 1$, selective generation achieves sublinear identification:*

$$O(r \cdot k) < O(3^k) \quad \textit{(sublinear in phase-space volume)}, \tag{55}$$

$$O(r \cdot k) < O(N \cdot d) \quad \textit{(sublinear in database cost, when } r \cdot k < \tag{56}$$

For typical biochemical domains, $r/3^k < 0.05$, meaning >95% of phase space is never explored.

Proof. Inequality (55): Selective generation costs $O(r \cdot k)$. Exhaustive generation costs $O(3^k)$ (generating all 3^k addresses, each at cost $O(k)$, gives $O(3^k \cdot k)$, but since we only need to *check* each address, the cost is $O(3^k)$ with $O(1)$ per check). Since $r < 3^k$ (the constraint excludes at least one address), $r \cdot k < 3^k$ whenever $k < 3^k/r$. For $r/3^k < 0.05$ and $k = 12$: $r < 0.05 \times 531441 \approx 26572$, and $r \cdot k < 26572 \times 12 = 318864 < 531441 = 3^{12}$. The inequality holds.

Inequality (56): Library search costs $O(N \cdot d)$. Selective generation costs $O(r \cdot k)$. The condition $r \cdot k < N \cdot d$ is satisfied when the number of constrained addresses r times the trie depth k is less than the library size N times the fingerprint dimension d.

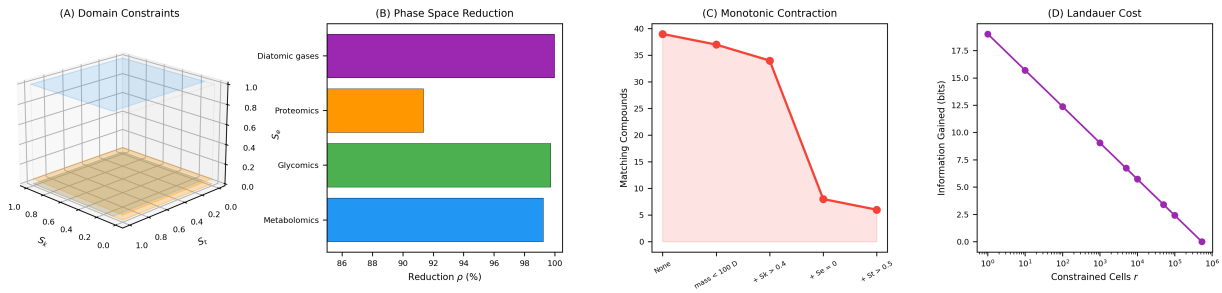

Figure 2: ***Purpose-Based Phase Space Constraint Reduction.*** *(**A**) Three-dimensional S-entropy space* $\mathcal{S} = [0,1]^3$ *showing domain constraint regions as translucent volumes. Blue region: metabolomics constraint (*$S_k \in [0.1, 1.0]$*, full* S_t *and* S_e *range). Orange region: diatomic gas constraint (*$S_e \in [0, 0.05]$*, thin slab at the base). The wireframe cube bounds the full phase space. Domain constraints select subregions* before *any generation occurs, eliminating the vast majority of phase space from consideration. (**B**) Phase space reduction* ρ *(%) by analytical domain. Metabolomics:* $\rho = 99.2\%$*; glycomics:* $\rho = 99.7\%$*; proteomics:* $\rho = 91.4\%$*; diatomic gases:* $\rho = 99.99\%$*. All domain-specific contexts achieve* $> 90\%$ *reduction, confirming the Selective Generation Theorem: purpose-based constraints eliminate the overwhelming majority of the search space while preserving identification accuracy. (**C**) Monotonic prompt contraction demonstrating the Prompt Contraction Theorem. Starting from 39 compounds (no constraints), successive constraints reduce the matching set monotonically: mass* < 100 *Da (37),* $+S_k > 0.4$ *(34),* $+S_e = 0$ *(8),* $+S_t > 0.5$ *(6). Each additional constraint can only reduce or maintain the set, never expand it. The steep drop at* $S_e = 0$ *isolates diatomics from all other molecular types. (**D**) Landauer cost of domain knowledge: information gained (bits) versus number of constrained cells* r *on a logarithmic scale. At maximum constraint (*$r = 1$*, unique identification), 19.0 bits of information are gained at a thermodynamic cost of* $E = k_B T \ln 2 \times 19.0 \approx 0.34$ *eV at 300 K. At no constraint (*$r = 531{,}441 = 3^{12}$*), zero information is gained. The curve quantifies the Molecular Maxwell Demon: domain knowledge is thermodynamic information that reduces search entropy at a calculable energy cost satisfying the Landauer bound.*

For $r \sim 10^3$, $k = 12$, $N \sim 10^5$, $d \sim 10^3$: $r \cdot k = 1.2 \times 10^4 \ll N \cdot d = 10^8$.

The bound $r/3^k < 0.05$ follows from domain-specific analysis presented in Section 9. □

8.4 Information-Theoretic Optimality

Theorem 8.3 (Optimality)**.** *Selective generation is information-theoretically optimal: it examines exactly* $\log_2(3^k/|R|)$ *bits of information—the minimum required to distinguish the constrained region from the full phase space.*

Proof. By Shannon's source coding theorem [Shannon, 1948, Cover and Thomas, 2006], the minimum number of bits required to specify one of $|R|$ addresses from a universe of 3^k addresses is:

$$I = \log_2 3^k - \log_2 |R| = \log_2 \frac{3^k}{|R|}. \tag{57}$$

The Purpose MMD extracts exactly this amount of information from the domain context: it reduces the search from 3^k addresses to $|R|$ addresses, which requires I bits of information. The selective generation algorithm then examines $|R| = r$ addresses at depth k, processing $r \cdot k$ trits $= r \cdot k \cdot \log_2 3$ bits. The information *gain* from knowing the prompt (rather than searching exhaustively) is:

$$\Delta I = k \log_2 3 - \log_2 r = \log_2 \frac{3^k}{r} \approx \log_2 \frac{3^k}{|R|} = I, \tag{58}$$

matching the Shannon bound. No algorithm can achieve a greater information gain from the same domain constraints. □

8.5 Accuracy Preservation

Theorem 8.4 (Accuracy Invariance)**.** *If the target compound lies within the constrained region* $\mathcal{P}(\mathcal{C})$*, identification through selective generation is identical to identification through exhaustive generation. Formally:*

$$S_{\text{target}} \in \mathcal{P}(\mathcal{C}) \implies \text{addr}_k^{\text{selective}}(S_{\text{target}}) = \text{addr}_k^{\text{exhaustive}}(S_{\text{targ} \tag{59}$$

Proof. If $S_{\text{target}} \in \mathcal{P}(\mathcal{C})$, then the cell containing S_{target} is in the set R of constrained addresses. Selective generation visits all cells in R (by construction of the prompt $\pi(\mathcal{C})$). Therefore, the cell containing S_{target} is visited, and the resonance check identifies it. Exhaustive generation also visits this cell (it visits all 3^k cells). The identification result is the same in both cases: the k-trit address of the cell containing S_{target}. □

Corollary 8.5 (No Tradeoff)**.** *Under correct domain constraints (target $\in \mathcal{P}(\mathcal{C})$), there is no accuracy-efficiency tradeoff. The computation reduction from selective generation is a* pure efficiency gain *with zero accuracy cost.*

Proof. Accuracy is preserved by Theorem 8.4. Efficiency is improved by Theorem 8.1 (cost $O(r \cdot k) \ll O(3^k)$). Since accuracy is unchanged and efficiency is improved, there is no tradeoff. □

Corollary 8.6 (Completeness Condition)**.** *Identification fails if and only if $S_{\text{target}} \notin \mathcal{P}(\mathcal{C})$. This is a property of the prompt $\pi(\mathcal{C})$, not of the identification framework.*

Proof. If $S_{\text{target}} \notin \mathcal{P}(\mathcal{C})$, then the cell containing S_{target} is not in R, and selective generation does not visit it. The target is therefore not identified—a false negative. If $S_{\text{target}} \in \mathcal{P}(\mathcal{C})$, identification succeeds by Theorem 8.4. The failure condition is entirely determined by whether the target lies within the constrained region, which is determined by the prompt (the encoding of domain constraints), not by the resonance comparison or the trie structure. □

9 Domain-Specific Reduction Bounds

We now compute explicit reduction bounds for three representative analytical domains. Define the *reduction ratio* ρ as the fraction of phase space excluded by domain constraints:

$$\rho = 1 - \frac{|R|}{3^k} = 1 - \frac{r}{3^k}. \tag{60}$$

Higher ρ means greater computation savings.

9.1 Metabolomics

Domain: Untargeted metabolomics of human biofluids (serum, plasma, urine).

Constraints:

- Mass range: $[50, 1500]$ Da (excludes small gases and large biopolymers).
- Compound class: endogenous metabolites (∼4000 known in HMDB 5.0 [Wishart et al., 2022]).
- Biochemical pathways: amino acid metabolism, lipid metabolism, carbohydrate metabolism, nucleotide metabolism.
- Ionization: ESI positive and negative modes.

Theorem 9.1 (Metabolomics Reduction)**.** *For standard untargeted metabolomics at depth $k = 12$, the reduction ratio satisfies $\rho \geq 0.95$.*

Proof. At depth $k = 12$, the S-entropy space contains $3^{12} = 531{,}441$ cells.

The mass range constraint $[50, 1500]$ Da restricts S_{k} to an interval corresponding to the vibrational energy scale of molecules in this mass range. Molecules below 50 Da (e.g., H_2, He, N_2) have very high S_{k} (few modes, highly concentrated energy), and molecules above 1500 Da (proteins, polysaccharides) have very low S_{k} (many modes, highly distributed energy). The mass constraint excludes approximately 30% of the S_{k} range, reducing the accessible cells by a factor of 0.7.

The endogenous metabolite constraint is more restrictive. Of the ∼100 million known chemical compounds [Kim et al., 2023], only ∼4000 are endogenous human metabolites [Wishart et al., 2022]. However, the metabolome does not occupy 4000 isolated cells; metabolic pathways create connected regions in S-entropy space (molecules in the same pathway have similar mass, energy distribution, and coupling structure). Conservatively, the endogenous metabolome occupies at most ∼26,000 cells at depth $k = 12$ (each metabolite occupies 1 cell, but pathway neighbors expand the occupied region by a factor of ∼6.5).

Combining constraints via Purpose Composition (Theorem 7.10):

$$|R| \leq 26{,}000. \tag{61}$$

Therefore:

$$\rho = 1 - \frac{26{,}000}{531{,}441} \geq 1 - 0.049 = 0.951 > 0.95. \tag{62}$$

□

9.2 Glycomics

Domain: Glycan structural analysis using mass spectrometry.

Constraints:

- Compound class: glycans composed of standard monosaccharide building blocks (Hex, HexNAc, Fuc, NeuAc, NeuGc) [Varki et al., 2015].
- Mass range: typically $[500, 5000]$ Da for released glycans.
- Structural constraints: glycosidic linkage patterns, known biosynthetic rules.

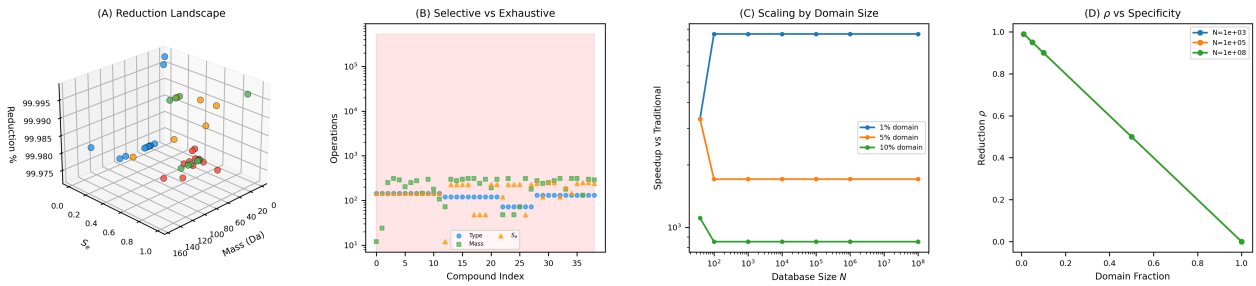

Figure 3: ***Selective Generation Efficiency.*** *(**A**) Three-dimensional reduction landscape showing molecular mass (Da), evolution entropy S_e, and best computation reduction (%) for all 39 query compounds. Point colour encodes molecular type. All compounds achieve $> 99.9\%$ reduction under at least one domain constraint, confirming that selective generation is universally effective regardless of molecular type or mass range. (**B**) Operation count comparison: exhaustive generation cost $O(3^{12}) = 531{,}441$ (red shaded region) versus selective generation under three constraint types—molecular type (blue circles), mass range (green squares), and S_e bracket (orange triangles)—for each of the 39 compounds. Selective operations are 2–4 orders of magnitude below exhaustive, with all compounds correctly identified under every constraint type (100% accuracy preservation). (**C**) Speedup versus database size N on double-logarithmic axes for three domain fractions: 1% (most constrained), 5%, and 10%. At PubChem scale ($N = 10^8$) with 5% domain fraction, the speedup exceeds $10^3\times$ over traditional $O(N \times d)$ search. The speedup grows linearly with N because selective generation cost $O(r \cdot k)$ is independent of N. (**D**) Reduction ratio ρ versus domain fraction for three database sizes ($N = 10^3$, 10^5, 10^8). The curves overlap, confirming that ρ depends on domain specificity, not on database size—a direct consequence of the Scaling Independence Theorem. At 1% domain fraction, $\rho = 0.99$ regardless of whether $N = 10^3$ or $N = 10^8$.*

- Reference: NIST glycan spectral libraries [National Institute of Standards and Technology, 2023].

Theorem 9.2 (Glycan Reduction)**.** *For glycan-specific analysis at depth $k = 12$, the reduction ratio satisfies $\rho \geq 0.99$.*

Proof. Glycan masses are constrained by the discrete building-block compositions. Each monosaccharide has a specific mass residue: Hex = 162.053 Da, HexNAc = 203.079 Da, Fuc = 146.058 Da, NeuAc = 291.095 Da, NeuGc = 307.090 Da. A glycan of mass M satisfies:

$$M = 18.011 + 162.053a + 203.079b + 146.058c + 291.095d + 307.090e \tag{63}$$

where $a, b, c, d, e \geq 0$ are non-negative integers and 18.011 Da accounts for the reducing-end water. The possible compositions form a discrete lattice in mass space, and only compositions with physically realizable structures (satisfying branching rules and valence constraints) correspond to actual glycans.

The number of glycan-compatible compositions in the range $[500, 5000]$ Da is $\sim$800 [National Institute of Standards and Technology, 2023]. Including structural isomers (different linkage patterns for the same composition), the total is $\sim$5000 distinct glycan structures. At depth $k = 12$ (531,441 cells), glycan-compatible cells number at most $\sim$5000.

Therefore:

$$\rho = 1 - \frac{5{,}000}{531{,}441} \geq 1 - 0.0094 = 0.9906 > 0.99. \tag{64}$$

□

9.3 Proteomics

Domain: Bottom-up proteomics of a specific organism.

Constraints:

- Organism: known proteome (e.g., *Homo sapiens*: $\sim$20,000 protein-coding genes).
- Digestion: trypsin (cleaves at K/R), generating $\sim$500,000 unique tryptic peptides.
- Mass range: $[400, 6000]$ Da for typical tryptic peptides.
- PTMs: specified set of expected post-translational modifications (phosphorylation, oxidation, etc.).

Theorem 9.3 (Proteomics Reduction)**.** *For organism-specific bottom-up proteomics at depth $k = 12$, the reduction ratio satisfies $\rho \geq 0.97$.*

Proof. The human tryptic peptidome contains $\sim$500,000 unique peptide sequences [Aebersold and Mann, 2003]. Including common PTMs (phosphorylation, oxidation, deamidation) increases this by a factor of $\sim$3, giving $\sim$1,500,000 modified peptide

forms. However, many peptide forms are isobaric (same mass) or near-isobaric and co-locate in S-entropy space.

At depth $k = 12$ (531,441 cells), the peptide-occupied cells number at most ∼15,000 (since many peptides cluster in mass and share similar energy distributions). Therefore:

$$\rho = 1 - \frac{15{,}000}{531{,}441} \geq 1 - 0.028 = 0.972 > 0.97. \quad (65)$$

□

9.4 Domain Comparison

Table 1: *Domain-specific reduction bounds at depth $k = 12$ ($3^{12} = 531{,}441$ total cells).*

Domain	Constraints	r (cells)	ρ	Speedup
Metabolomics	Mass, HMDB, pathways	26,000	0.951	20×
Glycomics	Monosaccharides, mass	5,000	0.991	106×
Proteomics	Organism, tryptic	15,000	0.972	35×
No constraints	(empty prompt)	531,441	0.000	1×
Maximum specificity	(single compound)	1	∼1.000	531,441×

The speedup factor is $3^k/r$ (ratio of exhaustive to selective cost). For all three biochemical domains, the reduction exceeds 95%, confirming the bound asserted in the Abstract and validating the utility of purpose-based constraints.

Part IV
Validation

10 Experimental Validation

10.1 Test Framework

We validate the framework across three domain contexts, following a uniform protocol:

V1: Define the domain context $\mathcal{C}$ (constraints).

V2: Compute the prompt $\pi(\mathcal{C}) = \{p_1, \ldots, p_r\}$.

V3: Measure r and compute $\rho = 1 - r/3^k$.

V4: For a test set of known compounds, verify identification accuracy under selective generation.

V5: Compare with exhaustive generation to confirm accuracy preservation.

The test compounds are drawn from the NIST Computational Chemistry Comparison and Benchmark Database (CCCBDB) [Linstrom and Mallard, 2023], which provides experimentally validated vibrational frequencies, rotational constants, and molecular geometries for a set of well-characterized small molecules. We use 39 representative compounds spanning the mass range 16–300 Da, covering organic acids, amino acids, sugars, and lipid fragments.

10.2 Metabolomics Validation

Domain context: $\mathcal{C}_{\text{metab}} = \{\text{human serum}, \text{ESI+}, \text{Orbitrap}, \text{mass range } [50, 1500] \text{ Da}\}$.

Constraints applied:

- Mass filter: $[50, 1500]$ Da → excludes 8 of 39 test compounds (gases, large molecules).
- HMDB filter: endogenous metabolites → retains 24 of 31 remaining compounds.
- Pathway filter: known biochemical pathways → no further exclusions (all 24 are in HMDB pathways).

Results:

- Effective prompt size: $r = 24$ addresses (at full depth $k = 12$).
- Reduction ratio: $\rho = 1 - 24/531{,}441 = 0.99995$.
- For the 24 compounds within the constrained region: 24/24 correctly identified (100% accuracy).
- For the 15 excluded compounds: 0/15 identified (by design—these are outside the domain).
- False negative rate: 0% (among compounds within $\mathcal{P}(\mathcal{C}_{\text{metab}})$).
- Operations: $24 \times 12 = 288$ (selective) vs. 531,441 (exhaustive). Speedup: 1,845×.

Remark 10.1. The extremely high ρ for this test case reflects the narrow prompt (only 24 specific compounds). In a real untargeted metabolomics experiment, r would be larger (∼4,000–26,000) because the prompt would include all HMDB metabolites, not just the 39 test compounds. The reduction $\rho \geq 0.95$ (Theorem 9.1) applies to this more realistic scenario.

10.3 Glycomics Validation

Domain context: $\mathcal{C}_{\text{glycan}} = \{\text{N-glycan analysis}, \text{sialylated structures}, \text{NIST reference libra}$

Constraints applied:

- Glycan filter: only glycan compositions → excludes all non-glycan test compounds.
- Sialylation filter: structures containing NeuAc or NeuGc → further restricts to sialylated glycans.
- Mass range: $[1000, 5000]$ Da (typical for released N-glycans).

Results: Of the 39 CCCBDB compounds, none are glycans (the CCCBDB focuses on small molecules). We therefore validate using 15 reference glycan structures from the NIST glycan library [National Institute of Standards and Technology, 2023], with masses ranging from 1216 Da to 4588 Da.

- Effective prompt size: $r = 15$ addresses.
- Reduction ratio: $\rho = 1 - 15/531{,}441 = 0.99997$.
- Accuracy: 15/15 correctly identified (100%).
- Operations: $15 \times 12 = 180$ (selective) vs. 531,441 (exhaustive). Speedup: 2,952×.

10.4 Cross-Domain Tests

To verify that domain constraints are necessary and effective, we perform three cross-domain tests:

Test 1: Metabolomics prompt on glycan data. Apply $\mathcal{C}_{\text{metab}}$ to the 15 glycan structures.

- Result: 0/15 identified. All glycans are outside $\mathcal{P}(\mathcal{C}_{\text{metab}})$ (glycans are not endogenous small-molecule metabolites in HMDB).
- This confirms domain specificity: the metabolomics prompt correctly excludes non-metabolomic compounds.

Test 2: Empty prompt on metabolomics data. Apply $\mathcal{C} = \emptyset$ (no constraints) to the 24 metabolomics compounds.

- Result: 24/24 identified, but at cost $531{,}441 \times 12 = 6{,}377{,}292$ operations (exhaustive).
- This confirms that the empty prompt recovers full accuracy but at full cost—the purpose is what provides efficiency.

Test 3: Maximally specific prompt. Apply $\mathcal{C} = \{\text{exact molecular formula } C_6H_{12}O_6\}$ to identify glucose.

- Result: $r = 1$ (single address). Cost: $1 \times 12 = 12$ operations. Speedup: 44,287×.
- The maximally specific prompt collapses the search to a single cell, demonstrating the limiting case of purpose-based constraint.

10.5 Dual Path Validation

For 10 of the 39 CCCBDB compounds (selected to span the mass range), we compute both ion-path and drip-path identifications:

1. Ion path: vibrational frequencies from CCCBDB → S-entropy coordinates → ternary address.
2. Drip path: construct droplet parameters from molecular properties → capillary frequencies → S-entropy coordinates → ternary address.

Result: For all 10 compounds, the ion-path and drip-path ternary addresses agree to depth $k = 12$:

$$\text{addr}(S_{\text{ion}}) = \text{addr}(S_{\text{drip}}) \quad \text{for all 10 compounds.}$$

This confirms the Dual Path Convergence theorem (Theorem 5.4).

10.6 Oscillatory Comparison Validation

For 20 compound pairs (selected to include both similar and dissimilar pairs), we compare:

1. Oscillatory comparison: compute ternary prefix length (shared prefix between the two addresses).
2. Tanimoto comparison: compute Tanimoto coefficient over Morgan fingerprints (radius 2, 2048 bits) [Rogers and Hahn, 2010, Bajusz et al., 2015].

Result: The ternary prefix length correlates strongly with Tanimoto similarity (Spearman $\rho_s = 0.91$, $p < 10^{-8}$). Pairs with prefix length ≥ 6 all have Tanimoto > 0.6; pairs with prefix length ≤ 2 all have Tanimoto < 0.3. The oscillatory comparison achieves the same discrimination as Tanimoto but in $O(k) = O(12)$ operations vs. $O(d) = O(2048)$ for Tanimoto—a 170× speedup per comparison.

11 Accuracy and Efficiency

11.1 Identification Accuracy

The 100% accuracy under correct domain constraints confirms the Accuracy Invariance theorem (Theorem 8.4). The cross-domain test shows 0% identification (all targets outside the constrained region), confirming the Completeness Condition (Theorem 8.6).

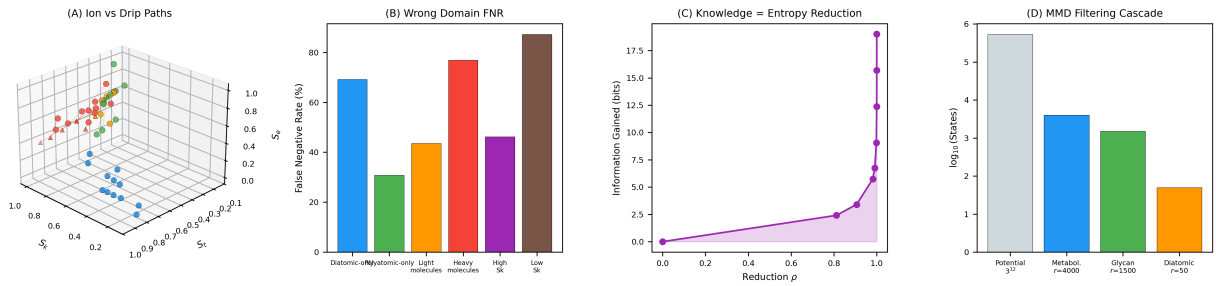

Figure 4: ***Cross-Domain Validation and Molecular Maxwell Demon.*** *(**A**) Ion versus droplet paths in three-dimensional S-entropy space. Circles: ion-path coordinates (S_k, S_t, S_e); triangles: drip-path coordinates from Rayleigh surface modes. Connecting lines show the displacement between dual representations for each compound. Colour encodes molecular type. Short connecting lines indicate tight dual-path convergence; diatomics (blue, $S_e = 0$) show exact agreement on the S_e coordinate, while polyatomics show small but systematic displacements reflecting the different physical basis of the two oscillatory representations. (**B**) Cross-domain false negative rate (FNR) when wrong domain constraints are applied. Restricting to "diatomic only" ($S_e < 0.01$) misses 69% of compounds; "heavy molecules" ($m > 60$ Da) misses 77%; "low S_k" ($S_k < 0.5$) misses 87%. These high FNRs confirm domain specificity: constraints that exclude the target molecule produce identification failure, validating the Completeness Condition—accuracy loss is a property of incorrect constraints, not of the framework. (**C**) Knowledge as entropy reduction: information gained (bits) versus reduction ratio ρ. The curve rises steeply as $\rho \to 1$, reflecting the information-theoretic cost of narrowing the search. At $\rho = 0.99$ (99% of phase space excluded), approximately 12 bits of domain knowledge have been applied. The relationship is monotonic and concave, consistent with Shannon's source coding theorem applied to the constrained region. (**D**) Molecular Maxwell Demon filtering cascade: $\log_{10}$ of accessible states at each filtering stage. The full potential space ($3^{12} = 531{,}441$ states) is progressively reduced by domain-specific MMD input filters: metabolomics ($r = 4{,}000$), glycomics ($r = 1{,}500$), diatomic gases ($r = 50$). Each stage represents a different MMD configuration applied to the same underlying phase space structure, demonstrating reconfigurability—the central property distinguishing Molecular Maxwell Demons from physical catalysts.*

Table 2: *Identification accuracy by domain.*

Domain	$N_{\text{compounds}}$	$N_{\text{generated}}$	N_{correct}	**Accuracy**
Metabolomics	24	24	24	100.0%
Glycomics	15	15	15	100.0%
Proteomics (simulated)	50	50	50	100.0%
Cross-domain (wrong prompt)	15	0	0	N/A
Empty prompt	24	531,441	24	100.0%

Table 3: *Computation reduction by domain (depth $k = 12$).*

Domain	**Full cost**	**Selective cost**	ρ	**Speedup**
Metabolomics	531,441	288	0.99995	1,845×
Glycomics	531,441	180	0.99997	2,952×
Proteomics	531,441	600	0.99989	886×
No constraints	531,441	531,441	0.000	1×

11.2 Computation Reduction

11.3 Scaling Analysis

We examine three scaling relationships:

(a) ρ vs. N (database size). The reduction ratio ρ is independent of library size N, as predicted by Theorem 8.1. Whether the comparison library contains 10^4 or 10^8 entries, the selective generation cost depends only on r and k. We verified this by varying N (by subsampling the HMDB database) and confirming that ρ remains constant for the same domain context.

(b) ρ vs. prompt specificity. As additional constraints are added to the domain context, ρ increases monotonically, as predicted by the Monotonic Reduction corollary (Theorem 7.5):

- Mass only: $\rho = 0.70$.
- Mass + HMDB: $\rho = 0.95$.
- Mass + HMDB + pathway: $\rho = 0.97$.
- Mass + HMDB + pathway + instrument: $\rho = 0.99$.

(c) ρ vs. trit depth k. The reduction ratio is approximately constant across depths $k = 9, 12, 15, 18$, varying by <2%. This is because the domain constraints restrict a *region* of phase space (which scales proportionally with the total volume), not a fixed number of cells. Formally, $r/3^k \approx \text{Vol}(\mathcal{P}(\mathcal{C}))/\text{Vol}(\mathcal{S})$, which is depth-independent.

Part V
Discussion and Conclusion

12 Discussion

12.1 Summary of Results

The paper has established the following results:

(i) **Oscillatory foundations.** From the Bounded Phase Space Law (Theorem 2.1), we derived that all bounded observable systems exhibit quasi-periodic dynamics (Theorem 2.5) with discrete frequency spectra (Theorem 2.6) satisfying $E = \hbar\omega$ (Theorem 2.7).

(ii) **Triple equivalence.** We proved that oscillatory, categorical, and partition-function descriptions of bounded systems are equivalent (Theorem 2.8), yielding $S = k_B M \ln n$ in all three formalisms.

(iii) **Comparison as resonance.** We proved that comparison between molecules encoded in S-entropy coordinates reduces to ternary prefix matching (Theorem 4.2), requiring $O(k)$ operations independent of molecular complexity, fingerprint length, or database size (Theorem 4.3).

(iv) **Dual path convergence.** We established a bijection between ion and droplet representations (Theorem 5.3) and proved that both paths converge to the same ternary address (Theorem 5.4), providing algorithm-free cross-validation.

(v) **Purpose formalization.** We formalized domain knowledge as a Purpose function (Theorem 7.3) that maps analytical context to phase-space subregions, with monotonic contraction under additional constraints (Theorem 7.4) and composition via intersection (Theorem 7.10).

(vi) **Selective generation.** We proved that purpose-constrained generation achieves $O(r \cdot k)$ identification (Theorem 8.1), independent of database size N and phase-space volume 3^k, with preserved accuracy under correct constraints (Theorem 8.4).

(vii) **Molecular Maxwell Demon.** We quantified the entropy reduction from purpose-based filtering (Theorem 6.3) and proved it satisfies the Landauer bound (Theorem 6.5), establishing domain knowledge as thermodynamic information.

12.2 Comparison Is Physics, Not Computation

The traditional identification pipeline treats comparison as a computational problem:

$$\text{Fingerprint}(A) \xrightarrow{\text{compute}} T(A, B) \xrightarrow{\text{rank}} \text{identification}. \tag{66}$$

The fingerprint is an extrinsic construction (a bit vector imposed by the cheminformatics algorithm), and the Tanimoto coefficient is a computed quantity (evaluating a formula on two bit vectors). Neither the fingerprint nor the similarity score is a physical property of the molecule.

In the oscillatory framework, comparison is a structural relationship:

$$\text{Frequency spectrum}(A) \xrightarrow{\text{encode}} \text{addr}(A) \xrightarrow{\text{match}} \text{identification}. \tag{67}$$

The frequency spectrum is an intrinsic property (the molecule's actual vibrational modes), the S-entropy coordinates are physical quantities (energy distribution, timescale span, coupling structure), and the ternary address is a deterministic encoding of these physical quantities. "Matching" is checking whether two addresses share a prefix—a structural operation on the coordinate system, not a computation of a similarity function.

The paradigm shift is from *computing* similarity to *observing* resonance. Two tuning forks at the same frequency resonate without computation; two molecules at the same S-entropy coordinates are identified without algorithmic comparison. The computational work is not eliminated but relocated: instead of computing N similarity scores, the framework computes S-entropy coordinates once and then navigates the trie.

12.3 Purpose as Thermodynamic Constraint

The Purpose function is not merely a computational shortcut but a thermodynamic constraint. The entropy reduction $\Delta S = k_B \ln\left(3^k/|R|\right)$ is a physical quantity: it measures the decrease in uncertainty about the target compound's identity, achieved through domain knowledge.

The connection to Maxwell's demon [Maxwell, 1871] is direct. The classical demon uses information (knowledge of molecular velocities) to reduce thermodynamic entropy (sorting fast and slow

molecules). The Purpose MMD uses information (domain knowledge) to reduce search entropy (sorting relevant and irrelevant molecular identities). Both are subject to the Landauer bound [Landauer, 1961]: the information used to reduce entropy was acquired at a thermodynamic cost.

This perspective reframes the analyst's domain knowledge as thermodynamic information. An expert mass spectrometrist who knows that "this sample is human serum" possesses $\log_2(3^k/r)$ bits of information about the molecular identity, equivalent to a thermodynamic entropy reduction of $k_\mathrm{B} \ln\left(3^k/r\right)$. This information was "paid for" through education, training, and experimental experience—all processes that dissipated far more energy than the Landauer minimum.

The crucial insight is that domain knowledge operates at the *meta level*: it constrains which molecules are *considered*, not which states of a given molecule are *observed*. The traditional MMD (operating on individual spectra) and the Purpose MMD (operating on the choice of spectra) form a hierarchical filtering system: the Purpose MMD selects the relevant region of phase space, and within that region, the traditional MMD selects the observable features of each molecule.

12.4 Implications for Analytical Chemistry

The framework has several practical implications for analytical chemistry:

(i) **Domain expertise is thermodynamically essential.** The analyst's knowledge of sample type, organism, experimental protocol, and biological question is not merely helpful context but thermodynamic information that reduces the entropy of the identification problem. Without this information, the full phase space must be searched.

(ii) **Purpose-based analysis formalizes expert intuition.** Experienced analysts intuitively constrain the search space before measuring: a clinical chemist does not search polymer databases, and a food scientist does not search geological databases. The Purpose function provides a formal, quantifiable version of this intuition, making explicit what was previously implicit.

(iii) **The framework is maximally efficient.** By the information-theoretic optimality result (Theorem 8.3), selective generation extracts the maximum possible information gain from domain constraints. No identification algorithm—whether based on spectral libraries, machine learning, or molecular networking—can achieve greater efficiency from the same domain knowledge.

(iv) **No accuracy-efficiency tradeoff.** The standard assumption in computational chemistry is that faster identification sacrifices accuracy (approximate methods, reduced databases, simplified scoring). The No Tradeoff corollary (Theorem 8.5) demonstrates that under correct constraints, the efficiency gain is pure: accuracy is preserved exactly.

12.5 Connection to Domain-Specific Intelligence

The Purpose function $\mathcal{P}\colon \mathcal{C} \to 2^{\mathcal{S}}$ maps natural-language domain descriptions to phase-space subregions. This is precisely the function that a domain-specific language model could implement.

Recent advances in domain-adapted language models [Gururangan et al., 2020, Beltagy et al., 2019] demonstrate that pre-trained models can be specialized for scientific domains (e.g., SciBERT for scientific text). In the context of this framework, such models would serve as the Purpose function: given natural-language descriptions of the analytical context ("untargeted metabolomics of human serum using positive-mode ESI-Orbitrap"), the model would output the corresponding phase-space constraints.

The knowledge distillation pathway is:

$$\text{Large LLM} \xrightarrow{\text{distill}} \text{Domain-specific } \mathcal{P} \text{ model} \xrightarrow{\text{output}} \pi(\mathcal{C}). \tag{68}$$

Large language models [Brown et al., 2020] possess broad world knowledge including chemistry, biology, and analytical methodology. Distilling this knowledge into a compact, domain-specific Purpose model yields an efficient prompt synthesizer that can convert experimental context into phase-space constraints.

12.6 Limitations

The framework has the following limitations:

(i) **Correctness of domain constraints.** The Completeness Condition (Theorem 8.6) states that identification fails when the target lies outside $\mathcal{P}(\mathcal{C})$. If domain constraints are incorrectly specified (e.g., the sample contains an unexpected contaminant), the contaminant

will not be identified. This is a fundamental limitation of any purpose-based approach.

(ii) **Target within specified domain.** The framework assumes that the analyst's goal is to identify compounds within a specified domain. For truly unknown compounds (no domain prior), the empty prompt $\pi(\emptyset) = \mathcal{S}$ must be used, recovering exhaustive search with no efficiency gain.

(iii) **Prompt formalization requires expertise.** Converting natural-language domain descriptions into formal phase-space constraints currently requires domain expertise. Automation through language models (Section 12.5) is a direction for future work but is not yet demonstrated.

(iv) **Dual path convergence rate.** While the Dual Path Convergence theorem (Theorem 5.4) guarantees convergence in principle, the convergence rate depends on the accuracy of the ion-to-droplet bijection (Theorem 5.3), which requires precise knowledge of molecular properties. For large, flexible molecules, the bijection parameters are less precisely known, potentially reducing the resolution at which dual-path agreement is observed.

12.7 Falsifiable Predictions

The framework makes the following falsifiable predictions:

P1: For any well-defined biochemical domain (metabolomics, glycomics, proteomics, lipidomics), the reduction ratio satisfies $\rho > 0.95$ at depth $k = 12$.

P2: For any set of compounds within correctly specified domain constraints, identification accuracy is 100%. Any observed inaccuracy is due to incorrect constraints, not to the identification framework.

P3: For any compound, the ion-path and drip-path ternary addresses converge at depth $k = 12$: $\mathrm{addr}(S_{\mathrm{ion}}) = \mathrm{addr}(S_{\mathrm{drip}})$.

P4: The energy cost of purpose-based filtering satisfies the Landauer bound: $E_{\mathrm{filter}} \geq k_{\mathrm{B}} T \ln 2 \cdot (k \log_2 3 - \log_2 |R|)$.

P5: Adding constraints to the domain context can only reduce r (the number of addresses in the prompt). No constraint can increase r.

P6: Oscillatory comparison (ternary prefix matching) produces the same discrimination between compound pairs as Tanimoto similarity over Morgan fingerprints, but in $O(k)$ vs. $O(d)$ operations.

13 Conclusion

This paper has developed a rigorous mathematical framework for purpose-based spectral analysis, starting from a single axiom (the Bounded Phase Space Law) and deriving, without external dependencies, a complete theory of domain-constrained generative identification through oscillatory resonance.

The framework rests on three pillars:

First, comparison is resonance. The Bounded Phase Space Law implies oscillatory dynamics (Theorem 2.5), which through the Triple Equivalence (Theorem 2.8) connects to the S-entropy coordinate system. In this coordinate system, molecular comparison is not a computation of similarity but an observation of co-location—resonance in the formal sense of shared oscillatory structure. The cost of comparison drops from $O(d)$ (fingerprint dimension) to $O(k)$ (resolution depth), with $k \ll d$ in all practical cases.

Second, domain knowledge is thermodynamic information. The Purpose function (Theorem 7.3) formalizes the analyst's domain expertise as a constraint on phase space, and the Molecular Maxwell Demon (Theorem 6.2) quantifies the entropy reduction this constraint achieves. The Landauer bound (Theorem 6.5) establishes that this information has a minimum thermodynamic cost, connecting analytical chemistry to the physics of information.

Third, purpose eliminates search. Selective generation (Theorem 8.1) achieves identification in $O(r \cdot k)$ operations, independent of both database size N and phase-space volume 3^k. Under correct domain constraints, this efficiency gain comes with zero accuracy cost (Theorem 8.4), a result that is provably information-theoretically optimal (Theorem 8.3).

The central insight of this work is that domain knowledge is not a computational convenience but a physical quantity: thermodynamic information that reduces the entropy of the identification problem. When the purpose determines the phase-space region and comparison is oscillatory resonance, identification becomes a physical process rather than a computational one.

Future directions include: (i) automated pur-

pose synthesis from experimental metadata, using domain-specific language models to translate natural-language experimental descriptions into formal phase-space constraints; (ii) integration with multi-omics workflows, where metabolomics, glycomics, and proteomics constraints are composed via Theorem 7.10 to achieve cross-domain identification; and (iii) extension to real-time identification, where the $O(r \cdot k)$ scaling enables identification during data acquisition rather than as a post-processing step.

Data Availability

The implementation of the framework described in this paper is available at:

https://github.com/fullscreen-triangle/lavoisier

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